“RailPlot Management System”

A

Project Report

Submitted in partial fulfillment of the requirement for the

award ofDegree of

Bachelor of Science in Information Technology

Submitted to

444 - Swatantrata Senani Shankarlal Agarwal Prabandhan Avum Takniki Mahavidyalay, Patna

Submitted By
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B.Sc.IT-3rd Year
Session-2023-2026

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Session 2023-2026

DATE:……

CERTIFICATE

This is to certify that the work embodies in this project entitled, "RailPlot Management System", being submitted by **<Pragya Kumari>** **(202344400170)** in partial fulfillment of the requirement for the award of "**Bachelor of Science in Information Technology**" to ***PATLIPUTRA UNIVERSITY, PATNA*** during the academic year 2023-26is a record of bonafide piece of work, carried out by him/her under our/my supervision and guidance in the "**Department of Computer Science & Applications", SSSAPTM, Patna.**

Dr. Neeraj Agrawal Director	**Dr. Neeraj Kumar** Dean- Academics	**Dr. Amit Kumar Shukla** Principal

Date:.....

APPROVAL CERTIFICATE

The project report entitled "RailPlot Management System", being submitted by **<Pragya Kumari>** (202344400170)has been examined by us and is hereby approved for the award of degree "**Bachelor of Science in Information Technology**", for which it has been submitted. It is understood that by this approval the undersigned do not necessarily endorse or approve any statement made, the opinion expressed or conclusion drawn therein, but approve the project only for the purpose for which it has been submitted.

(Internal Examiner) (External Examiner)

Date: Date:

Date:.....

DECLARATION

I, <Pragya Kumari> **hereby declare that the work, which is being presented in the project report entitled**"**RailPlot Management System**",
partial fulfillment of the requirements for the award of degree of "**Bachelor of Science in Information Technology**" submitted in the department of **Computer Science & Applications (SSSAPTM)** is an authentic record of my own work carried under the joint guidance of **Prof. Niraj Kumar Singh Sir and Prof. Anjesh Kumar Sir.** I have not submitted the matter embodied in this report for the award of any other degree.

PragyaKumari

RegistrationNO.202344400170

B.Sc.IT 3rd Year

Session-2023-2026

ACKNOWLEDGEMENT

"A journey is easier when you travel together. Interdependence is certainly more valuable than independence."

I would like to thanks, **Prof. Niraj Kumar Singh Sir** and **Prof. Anjesh Kumar Sir**, for providing regular guidance and insight into my project work. I also thank them for all advice he has givenme in the past year, and for always having time for me, whenever I needed.

I give special thanks to **Dr. Amit Kumar Shukla Sir, Prof. Raju Upadhyay Sir,** and **Prof. Ravi Kumar Soni Sir** for always being willing to help find solutions to any problems I had with my work.

"The completion of any project depends upon the cooperation, coordination, and combined efforts of several resources of knowledge, inspiration, and energy".

I also extend my deepest gratitude to **Director**, CIMAGE GROUP OF INSTITUITIONS, PATNA, **Dr. Neeraj Agrawal Sir, Dean** CIMAGE GROUP OF INSTITUITIONS, PATNA, **Dr. Neeraj Kumar Sir** and **Centre Head Mrs. Megha Agrawal Ma'am** for providing all the necessary facilities and true encouraging environment to bring out the best in my endeavors.

I express my gratitude and thanks to all the staff members of Computer Science departments for their sincere cooperation in furnishing relevant information to complete this Project well in time successfully.

I extend a special word to my friends, who have been a constant source of inspiration throughout my project work.

Lastly but not least I must express my cordial thank to my parent and family members who gave me the moral support without that it is impossible to complete my project work. With this note, I thank everyone for the support.

PragyaKumari

RegistrationNO.202344400170

B.Sc.IT 3rd Year

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CONTENTS

INTRODUCTION 8

OBJECTIVE 9

PROJECT CATEGORY WEB APPLICATION 10

ADVANTAGES OF WEB APPLICATION 11

TOOLS/PLATFORM required 13

HARDWARE & SOFTWARE REQUIREMENTS 13

PROBLEM DEFINITION 14

REQUIREMENT SPECIFICATIONS 14

PROJECT PLANNING & SCHEDULING 14

GANTT CHART 16

PERT CHART 18

SYSTEM ANALYSIS 1

IDENTIFICATION OF NEED 1

FEASIBILITY STUDY 6

TECHNICAL FEASIBILITY: 7

ECONOMICAL FEASIBILITY: 7

OPERATIONAL FEASIBILITY: 8

ANALYSIS 10

DFD 10

MODULE DESCRIPTION 16

DATA STRUCTURE Snap shot 17

LIST OF REPORTS 22

SCOPE AND FUTURE ENHANCEMENT 23

IMPLEMENTATION METHODOLOGY 23

TESTING 25

LIMITATIONS AND FURTHER ENHANCEMENTS 28

SECURITY MECHANISM 29

SNAPSHOTS 30

“RailPlot Management System”

INTRODUCTION

The “RAILPLOT SYSTEM” has been designed and developed using modern software development technologies, primarily focusing on a Java-based backend system along with API architecture and database integration. The system is structured in such a way that it efficiently manages railway-related data and provides seamless communication between different modules through well-defined APIs.

In today’s digital era, backend technologies play a crucial role in building scalable and efficient applications. The backend of the RailPlot System is developed using advanced programming concepts that ensure high performance, reliability, and the ability to handle multiple user requests simultaneously. The use of API-based architecture makes the system flexible and allows it to be easily integrated with other platforms such as web applications or mobile applications.

The system also incorporates a database management system (such as MySQL or similar relational database) in the background, which is responsible for storing and managing large volumes of railway data. The database provides a structured query mechanism that

allows efficient data retrieval, insertion, and updating. It ensures data consistency, integrity, and security by restricting unauthorized access and maintaining proper data relationships.

One of the key features of the RailPlot System is its ability to organize and visualize railway-related information effectively, making it easier for users or administrators to access and analyze data. The system is designed keeping in mind the need for accuracy, speed, and security, which are essential in handling real-time data operations.

The development of the RailPlot System involved careful analysis of requirements and implementation of suitable technologies to meet those needs. Special attention has been given to system performance, data protection, and user accessibility, ensuring that the application remains reliable under different working conditions.

Overall, the “RAILPLOT SYSTEM” provides a powerful and efficient solution for managing railway data through a secure and well-structured environment. The combination of modern backend technologies, API-driven communication, and database support makes the system highly effective, scalable, and suitable for real-world applications.

OBJECTIVE

The motive of the project entitled **"RailPlot System"** is to efficiently manage, process, and visualize railway-related data using a structured and API-based system. The system is designed to handle backend operations smoothly while ensuring accurate data flow and secure data management.

The objectives of this system are:

• To manage and organize railway-related data in a structured manner.

• To provide efficient APIs for handling data operations such as insertion, retrieval, and updating of records.

• To ensure secure storage and management of data using a database system.

• To enable smooth communication between different components of the system through API architecture.

• To improve the accuracy and speed of data processing.

• To maintain proper system performance even with multiple user requests.

• To provide easy access to relevant data for analysis and visualization purposes.

• To ensure data consistency and integrity throughout the system.

• To maintain proper backup and recovery mechanisms for data safety.

• To make the system scalable so that it can be enhanced in the future with additional features.

PROJECT CATEGORY WEB APPLICATION

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The project **"RailPlot System"** falls under the category of a **Web Application**, which is designed to operate over a network and provide services through web-based technologies. A web application is a software system that runs on a server and can be accessed by users through web browsers or other connected platforms.

Web applications are widely used in modern software development due to their **flexibility, scalability, and ease of access**. Unlike traditional desktop applications, they do not require installation on individual systems and can be accessed from anywhere using an internet connection. This makes them highly efficient for real-time data processing and multi-user environments.

The RailPlot System is developed using a **backend API architecture**, which acts as the core component of the web application. It processes user requests, manages data operations, and communicates with the database. This structure allows the system to handle multiple requests simultaneously and ensures smooth and efficient performance.

The application also uses a **database management system** to store and manage railway-related data securely. The database ensures proper organization of data and provides features like **data integrity,**

consistency, and security, which are essential for any web-based system.

Another important advantage of web applications is their ability to **integrate with different platforms**, such as web interfaces or mobile applications. The RailPlot System is designed in such a way that it can be easily extended or connected with other systems in the future.

Overall, the **RailPlot System** as a web application provides a **reliable, scalable, and efficient solution** for handling railway-related data through a secure and structured environment.

ADVANTAGES OF WEB APPLICATION

1. Accessibility:

   Web applications can be accessed from anywhere using a web browser and internet connection. This makes the system highly convenient for users as they are not restricted to a specific device or location.

2. No Installation Required:

Unlike desktop applications, web applications do not require installation on individual systems. Users can directly access the application through a browser, saving time and system resources.

3. Platform Independence:

Web applications are platform-independent, meaning they can run on any operating system such as Windows, Linux, or macOS without compatibility issues.

4. Centralized Data Management:

All data is stored on a central server or database, which ensures better data management, consistency, and easy access. It also simplifies data backup and recovery processes.

5. Scalability:

   Web applications can be easily scaled to handle increasing numbers of users or data. This makes the system suitable for future expansion and improvements.

6. Easy Maintenance and Updates:

   Updates and changes can be made on the server side without requiring users to update anything on their devices. This ensures that all users always access the latest version of the application.

7. Security:

   Web applications provide better security as data is stored on secure servers. Proper authentication and authorization mechanisms can be implemented to protect sensitive information.

8. Integration Capability:

Web applications can be easily integrated with other systems, APIs, or services, making them flexible and suitable for modern software environments.

9. Multi-user Support:
   Multiple users can access the system simultaneously, making it efficient for real-time applications and collaborative environments.
10. Cost Effective:
    Since there is no need for installation or high-end hardware, web applications reduce overall costs in terms of deployment, maintenance, and usage.

TOOLS/PLATFORM REQUIRED

Operating System

- • Windows 10 / Windows 11
- • Linux-based Operating Systems
- • macOS

HARDWARE & SOFTWARE REQUIREMENTS

Hardware Requirements:

• Processor: Intel i3 or above
• RAM: Minimum 4 GB
• Hard Disk: Minimum 20 GB free space
• Internet Connection

Software Requirements:

• Operating System: Windows 10/11, Linux, or macOS
• Web Browser: Google Chrome / Mozilla Firefox / Microsoft Edge
• Backend Environment: (e.g., Node.js / Python / API framework)
• Database: MySQL or any relational database
• Code Editor: Visual Studio Code / Sublime Text / Notepad++

PROBLEM DEFINITION

The existing system for managing railway-related data is often **manual or not well-organized**, which leads to difficulties in handling large amounts of information efficiently. Managing data without a proper system can result in **data inconsistency, delays in processing, and lack of real-time access**.

There is a need for a system that can **efficiently store, manage, and process railway-related data** in a structured manner. Without such a system, it becomes difficult to track information, perform operations quickly, and ensure data accuracy.

The problem also includes the lack of a **centralized platform** where all the required data can be accessed and managed securely. Traditional methods do not support **multiple users, fast data retrieval, or integration with other systems**, which limits overall performance and usability.

Therefore, the RailPlot System is developed to overcome these issues by providing a **web-based solution** that ensures efficient data handling, secure storage, and easy accessibility through API-based architecture.

REQUIREMENT SPECIFICATIONS

A **Software Requirements Specification (SRS)** is a detailed description of the system to be developed. It defines the functional and non-functional requirements of the system and describes how users will interact with it. Functional requirements specify the operations and features of the system, while non-functional requirements define constraints such as performance, security, and reliability.

In the **RailPlot System**, the SRS provides a clear understanding of how the system will manage railway-related data through a web-based API structure. It ensures that all system functionalities are properly defined and implemented.

General Requirements of the Project:

• To manage and organize railway-related data in a structured format.

• To provide APIs for handling operations such as data insertion, retrieval, updating, and deletion.

• To maintain accurate records of system data for efficient processing and analysis.

• To ensure secure access to the system through proper authentication and authorization mechanisms.

• To support multiple users and handle concurrent requests efficiently.

• To maintain backup of important data to prevent data loss.

• To ensure fast and reliable data processing for better system performance.

• To provide easy integration with other applications or platforms if required.

PROJECT PLANNING & SCHEDULING

Project planning is an essential part of project management, which involves organizing tasks, resources, and timelines to ensure successful completion of the project. It includes the use of planning tools such as **Gantt charts** and scheduling techniques to track progress and manage different phases of the project effectively.

In the **RailPlot System**, the project planning phase began with defining the **project scope**, which includes developing a web-based application for managing and processing railway-related data through APIs. After identifying the scope, suitable methods and technologies were selected for implementation.

The project was then divided into smaller tasks using a **Work Breakdown Structure (WBS)**. These tasks included requirement analysis, system design, backend/API development, database integration, testing, and deployment. Each task was assigned an estimated time duration to ensure proper scheduling.

The logical sequence of tasks was determined to maintain smooth workflow, and dependencies between tasks were identified. This helped in understanding the **critical path**, ensuring that important

tasks were completed on time without delays. Resource allocation was also considered, including development tools, system requirements, and time management.

The project schedule was prepared to map tasks with timelines and resources. This schedule acted as a **baseline**, against which the actual progress of the project was measured. Regular monitoring and adjustments were made to ensure that the project stayed on track and met its objectives.

Proper planning helped in maintaining **efficiency, reducing errors, and ensuring timely completion** of the RailPlot System. It also provided clarity in development stages and improved overall project management.

GANTT CHART

Gantt charts, developed by Henry L. Gantt, are a project control technique used for scheduling, budgeting, and resource planning. A Gantt chart is a type of bar chart in which each bar represents a specific activity of the project.

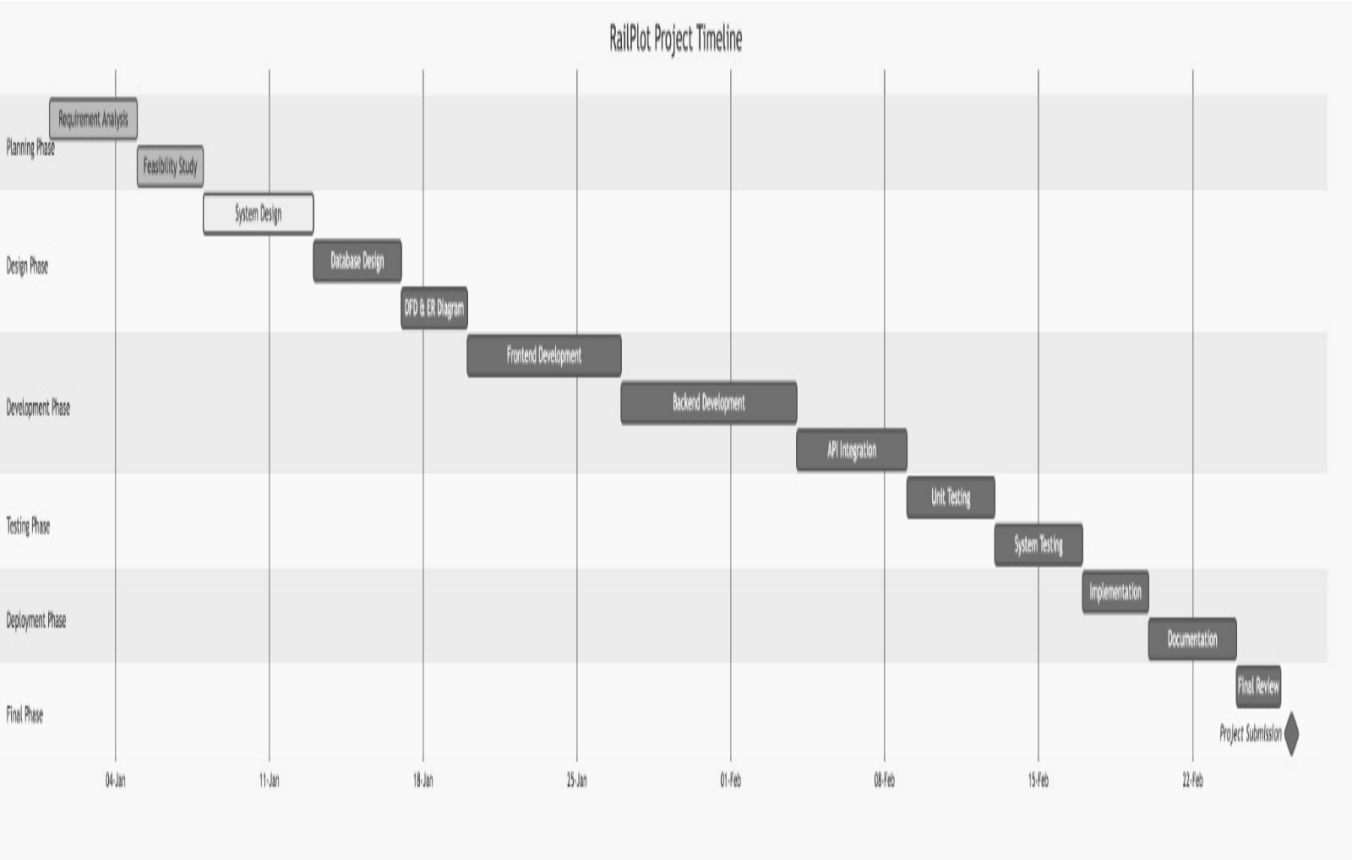
The bars are drawn against a timeline, where the length of each bar is proportional to the time allocated for that activity. This helps in understanding the sequence of tasks, tracking progress, and ensuring that the project is completed within the scheduled time.

In the **RailPlot System**, the Gantt chart is used for effective time management and planning of different development phases. The project is divided into multiple stages such as requirement analysis, system design, API/backend development, database integration, testing, and deployment. Each phase is assigned a specific time duration to ensure smooth execution of the project.

The Gantt chart helps in identifying task dependencies and ensures that all activities are completed in a logical sequence. It also allows monitoring of progress and helps in making necessary adjustments during the development process.

Thus, the Gantt chart of the **RailPlot System** is prepared to manage project time efficiently and to ensure timely completion of all phases. The Gantt chart is shown below:

VSimple Table Version (If teacher prefers)
PhaseTask Duration
Planning Requirement Analysis 5 Days
Planning Feasibility Study 4 Days
Design System Design6 Days
Design Database Design 5 Days
Design DFD & ER Diagram 4 Days
Development Frontend 8 Days
Development Backend 10 Days
Development API Integration 6 Days
Testing Unit Testing 5 Days
Testing System Testing 5 Days
Deployment Implementation 4 Days
Deployment Documentation 5 Days
Final Review & Submission 4 Days



PERT CHART

PERT (Project Evaluation and Review Technique) charts consist of a network of boxes and arrows, where the boxes represent activities and the arrows indicate the dependencies between tasks. It is an important project management tool used for planning, scheduling, and coordinating tasks in a project.

PERT is organized based on **events and activities**, and it is especially useful for managing complex projects. It helps in identifying different paths of task execution and analyzing the sequence in which activities must be completed.

In the **RailPlot System**, the PERT chart is used to define various stages of development such as requirement analysis, system design, backend/API development, database integration, testing, and deployment. Each of these tasks is interconnected, and the completion of one task may depend on the completion of another.

The PERT technique enables the calculation of the **critical path**, which is the sequence of activities that requires the maximum time to complete the project. Identifying the critical path helps the project manager to focus on important tasks and ensure that there are no delays in the overall project timeline.

Additionally, PERT helps in estimating the **time and cost associated with each task**, allowing better control over project resources. It also provides flexibility in managing uncertainties, as multiple task paths can be analyzed and adjusted when required.

Thus, the PERT chart of the **RailPlot System** helps in effective project planning, time management, and resource allocation, ensuring that the project is completed efficiently and within the defined schedule. The PERT chart representation of the system is shown below:

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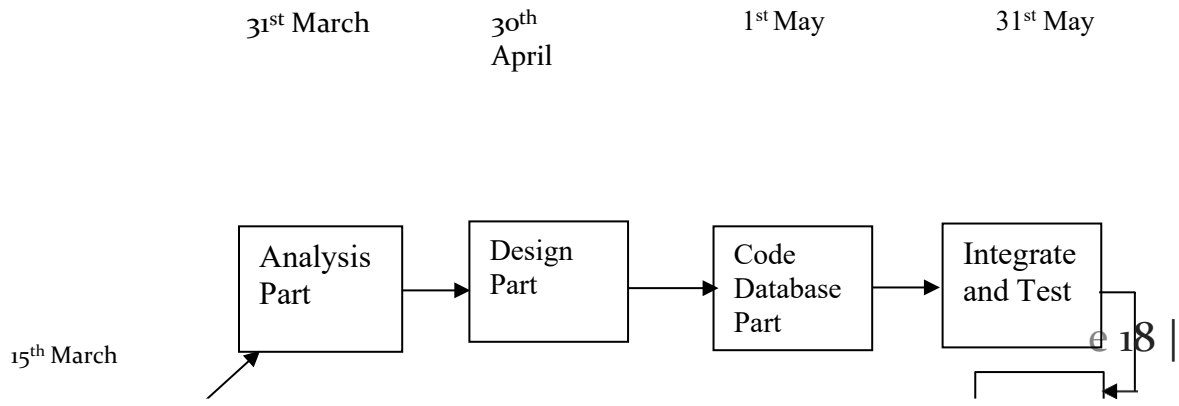


SYSTEM ANALYSIS

IDENTIFICATION OF NEED

For the development of the **RailPlot System**, the principles of **Requirement Engineering** are used to understand the needs of the system. Requirement engineering provides a structured approach to identify user requirements, analyze them, assess feasibility, and convert them into a functional system.

The requirement engineering process includes the following steps:

- Requirement Elicitation
- Requirement Analysis and Negotiation
- Requirement Specification
- System Modeling
- Requirement Validation
- Requirement Management

Requirement analysis acts as a bridge between system-level requirements and software design. It helps in understanding the **data, functional, and behavioral aspects** of the system. This allows developers to design the system in a way that meets user needs effectively.

In the **RailPlot System**, requirement analysis helps in defining how railway-related data will be managed, processed, and accessed through APIs. It ensures that the system is designed with proper structure, performance, and usability.

The most commonly used technique for gathering requirements is **interaction with users and stakeholders**. During this phase, basic questions are asked to understand the system clearly.

TYPES OF QUESTION CONSIDERD :

1. Basic Understanding Questions:

• Who will use the system?
• What is the main purpose of the system?
• What benefits will the system provide?

2. Problem Understanding Questions:

• What problems will the system solve?
• What type of output is expected?
• In which environment will the system be used?
• Are there any performance or technical constraints?

3. Validation Questions:

• Are the requirements correct and complete?
• Can additional information be provided?
• Are all necessary features included?

Based on these interactions and analysis, the requirements of the RailPlot System are clearly defined and structured.

PRELIMINARY INVESTIGATION

The preliminary investigation is the first step in the **System Development Life Cycle (SDLC)**. It is conducted to determine the feasibility and scope of the project. The purpose is not to design the system in detail, but to collect enough information to evaluate whether the project should proceed.

For the **RailPlot System**, the preliminary investigation involved the following objectives:

• To clearly understand the project requirements
• To determine the size and scope of the project

- To analyze possible approaches for development
- To assess the technical and operational feasibility
- To identify expected benefits of the system
- To ensure that the system meets user requirements

The data for this project was gathered through:

- Studying similar systems and applications
- Observing existing data handling methods
- Analyzing system requirements
- Reviewing available technical resources

Based on the investigation, it was concluded that developing a **web-based RailPlot System** is feasible and beneficial. The system provides structured, secure, and efficient way to manage and process railway-related data.

FEASIBILITY STUDY

Not every project idea is practically possible to implement, even in software development. A **feasibility study** is conducted to determine whether a proposed system is viable and worth developing. It helps in identifying whether the project can be completed successfully within given constraints such as time, cost, and available resources.

For the **RailPlot System**, the feasibility study is carried out across the following seven dimensions:

TECHNICAL FEASIBILITY: -

Technical feasibility evaluates whether the required technology, tools, and resources are available to develop the system.

The **RailPlot System** is technically feasible because:

- It is developed using modern web technologies (Frontend + Backend APIs)
- Required tools and platforms are easily available
- The system can run on standard operating systems like Windows
- No advanced or unavailable technology is required

ECONOMICAL FEASIBILITY: -

ts by imEconomic feasibility determines whether the project is cost-effective.

- Development cost is low (uses open-source tools)
- No expensive hardware is required
- Reduces operational and manual costs
- Provides long-term benefiproving efficiency

OPERATIONAL FEASIBILITY: -

Operational feasibility checks whether the system will function properly in the real-world environment and be accepted by users.

- The system is user-friendly and easy to operate
- It reduces manual work in managing railway-related data
- Users can easily interact with the system through a web interface
- Requires minimal training

ANALYSIS

System Analysis refers to the process of studying and examining an existing situation with the aim of improving it through better procedures, methods, and systems. It involves understanding the current system, identifying problems, and proposing an improved solution.

System Analysis is also the process of planning and designing a new system to either replace or enhance an existing system.

In the case of the **RailPlot System**, system analysis focuses on understanding how railway-related data is currently managed and identifying the limitations of manual or less efficient systems. The analysis helps in designing a web-based system that provides better data management, faster processing, and improved accessibility.

The main objectives of system analysis in this project are:

- To understand the existing system and its drawbacks
- To identify user requirements and system needs
- To improve efficiency and reduce manual effort
- To design a structured and reliable system
- To ensure better performance and user satisfaction

A **Data Flow Diagram (DFD)** is an important tool used in system analysis to represent the flow of data within a system. It helps analysts to model the system's processes, data storage, and data movement in a simple and understandable way.

The DFD is an excellent communication tool for system analysts, as it clearly shows how data is processed and transferred between different components of the system. When used effectively, it becomes an easy-to-understand and powerful modeling technique.

For the **RailPlot System**, the DFD helps in visualizing how railway-related data flows between users, system processes, and databases. It provides a clear picture of system functionality and interactions.

Key Components of DFD

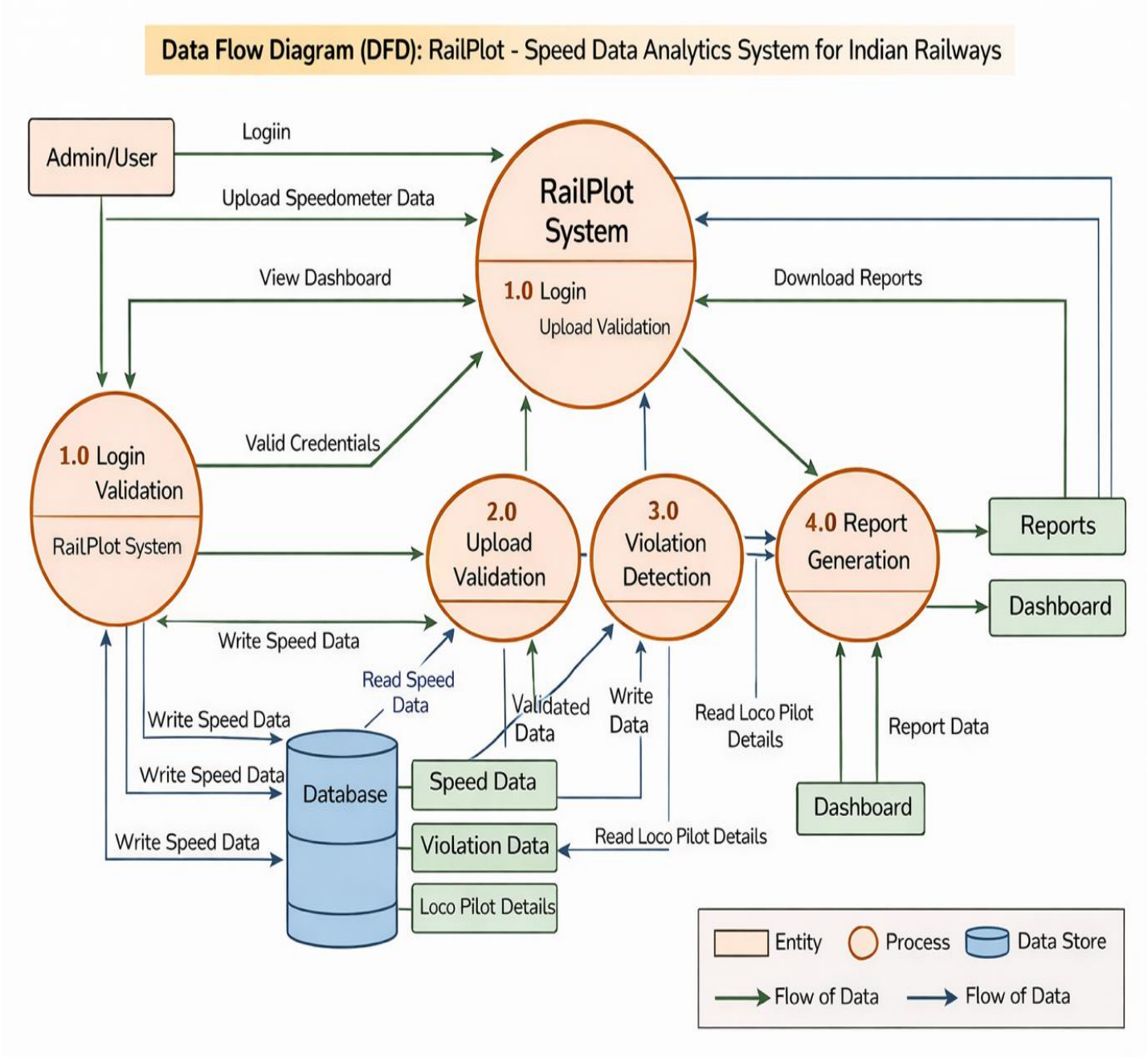
- **Processes** – Represent operations or activities performed on data
- **Data Flow** – Shows the movement of data between entities
- **Data Store** – Represents storage locations like databases
- **External Entities** – Users or external systems interacting with the system

Advantages of DFD

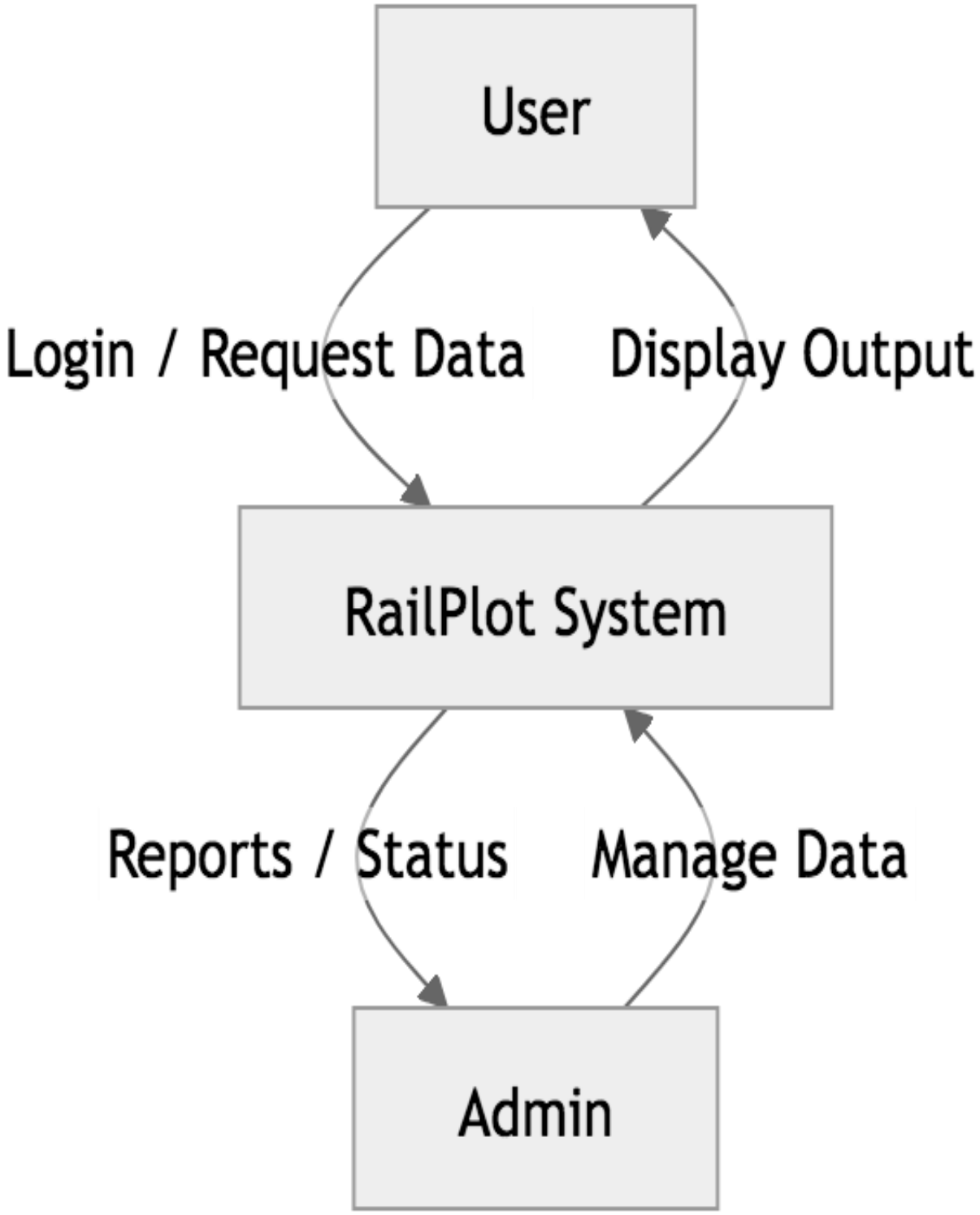
- Easy to understand and interpret
- Helps in effective communication between users and

developers

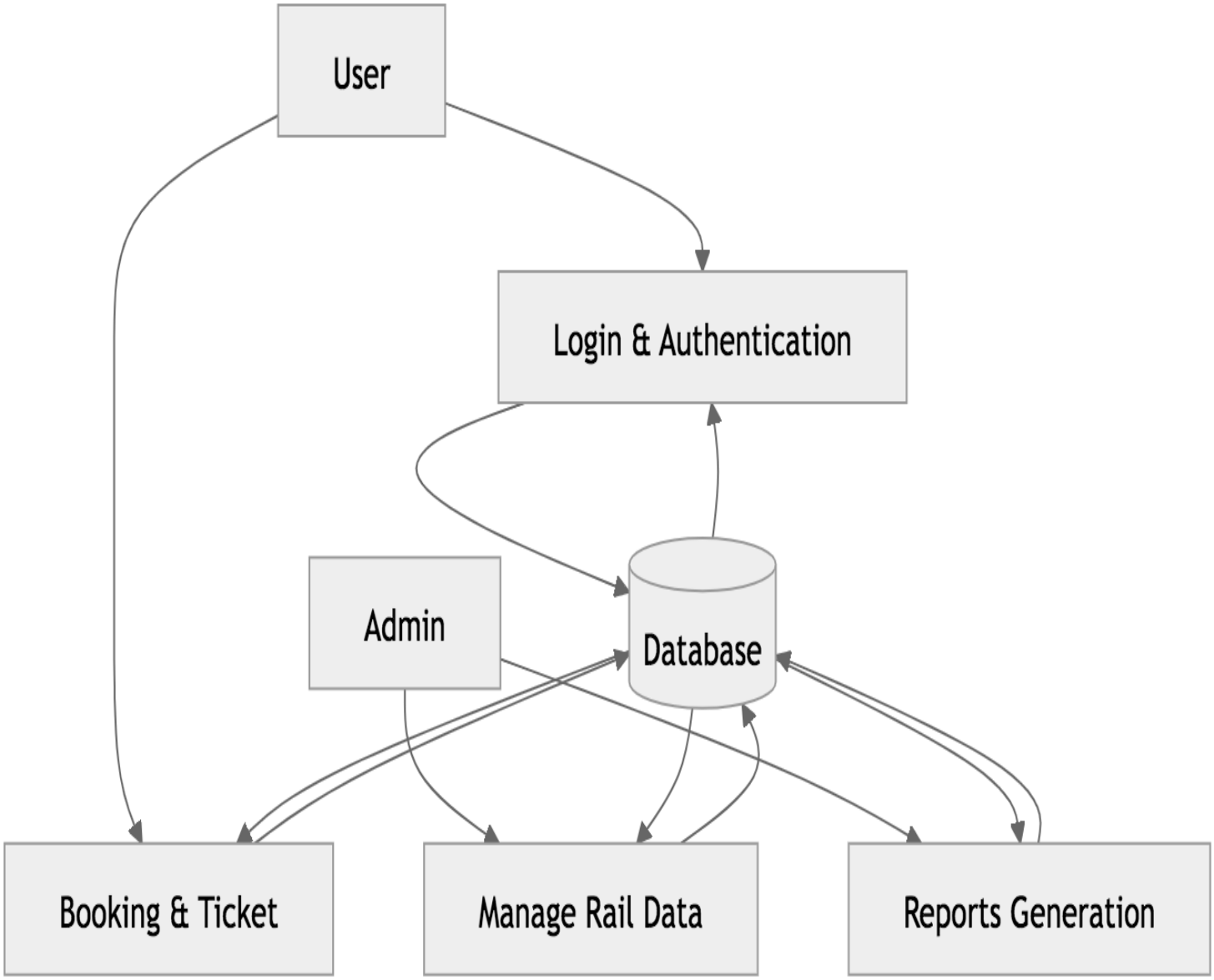
- Provides a clear overview of system processes
- Useful for system design and documentation
- Identifies data flow and system boundaries


Data Flow Diagram (DFD): RailPlot - Speed Data Analytics System for Indian Railways
Admin/User
Logiin
Upload Speedometer Data
View Dashboard
RailPlot System
1.0 Login
Upload Validation
Download Reports
1.0 Login Validation
RailPlot System
Valid Credentials
2.0 Upload Validation
3.0 Violation Detection
4.0 Report Generation
Reports
Dashboard
Write Speed Data
Read Speed Data
Validated Data
Write Data
Read Loco Pilot Details
Report Data
Write Speed Data
Write Speed Data
Write Speed Data
Database
Speed Data
Violation Data
Loco Pilot Details
Read Loco Pilot Details
Dashboard
Entity
Process
Data Store
Flow of Data
Flow of Data

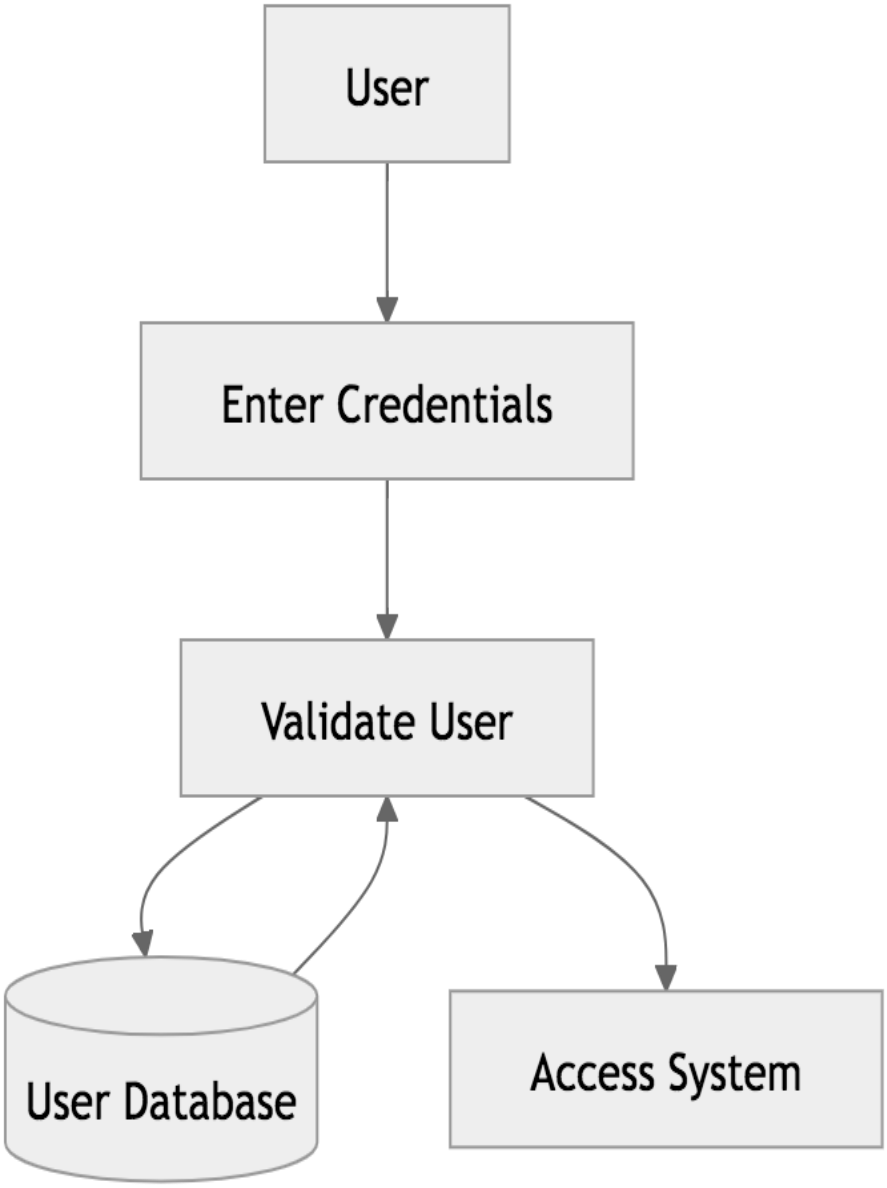
DFD LEVEL 0 (Context Diagram)



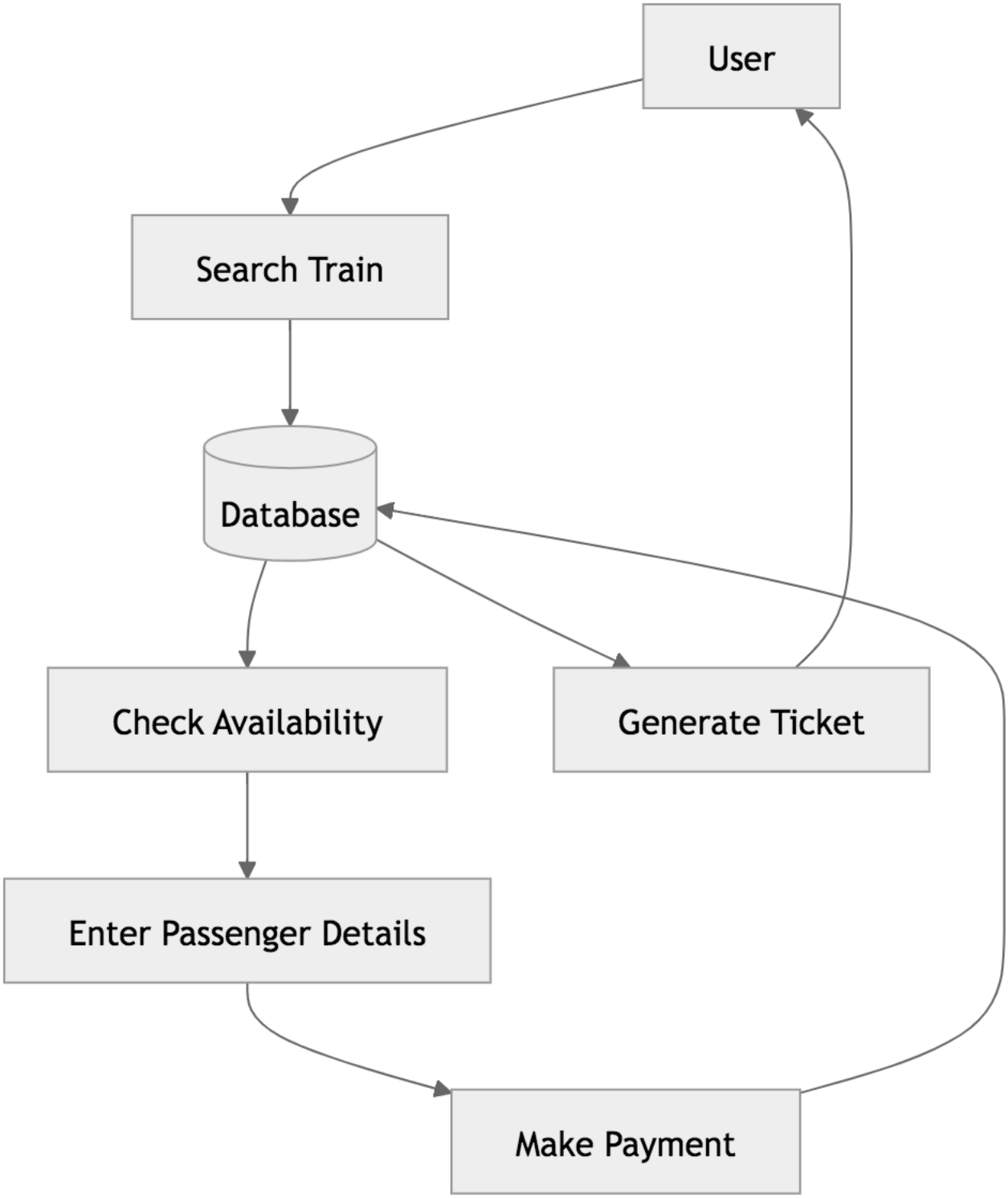
DFD LEVEL 1


User
Login & Authentication
Admin
Database
Booking & Ticket
Manage Rail Data
Reports Generation

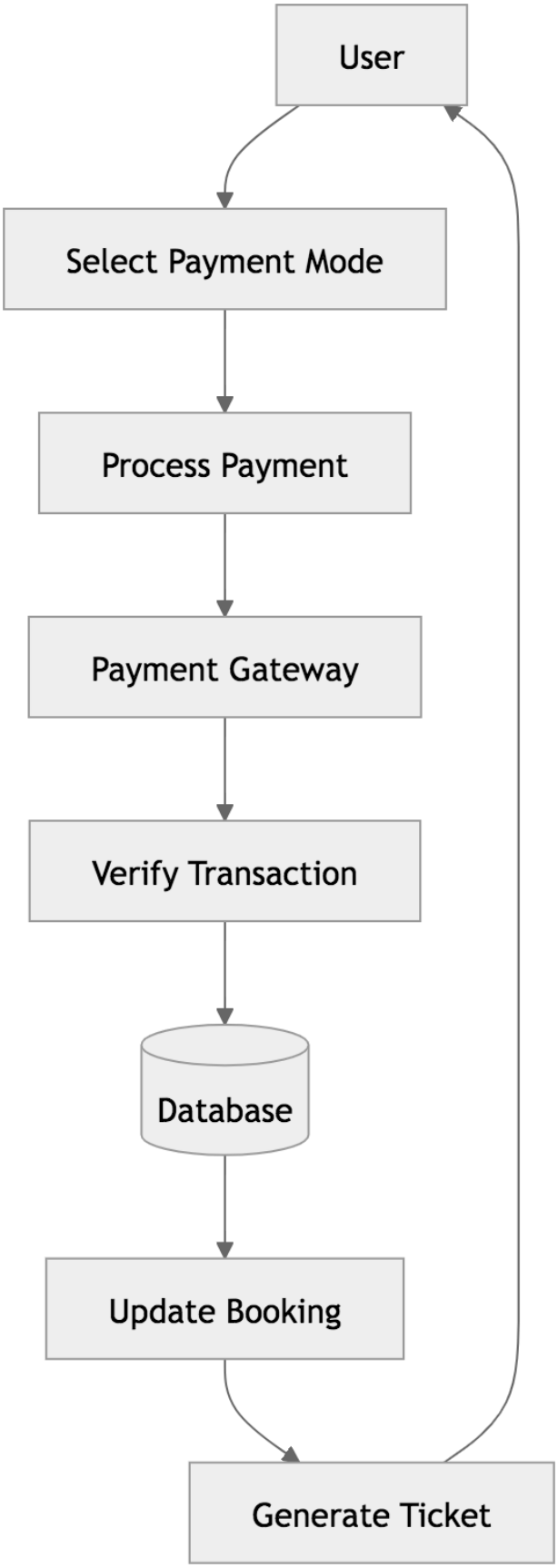
DFD LEVEL 2 (Detailed Process)


User
Enter Credentials
Validate User
User Database
Access System

DFD LEVEL 3 (Booking Process)



DFD LEVEL 4 (Payment & Ticket System)


User
Select Payment Mode
Process Payment
Payment Gateway
Verify Transaction
Database
Update Booking
Generate Ticket

MODULE DESCRIPTION

The **RailPlot System** is divided into different modules to ensure proper organization, efficient data handling, and smooth functioning of the application.

1 USER MODULE

This module manages the details of users who interact with the system. It stores user information such as name, role, and contact details.

- User registration and login
- Managing user profiles
- Role-based access (admin/user)

2. DATA MANAGEMENT MODULE

This module is responsible for handling railway-related data within the system.

- Add, update, and delete railway data
- Store and manage records efficiently
- Ensure data accuracy and consistency

3. API MODULE

This module provides communication between frontend and backend using APIs.

- Handles data requests and responses
- Ensures smooth data transfer
- Supports integration with other systems

4. AUTHENTICATION & SECURITY MODULE

This module ensures system security by managing user authentication.

- Login using username and password
- Secure access control
- Protects data from unauthorized access

5 ·

This module is used to generate and view reports based on system data.

• View system data in structured format
• Generate reports for analysis
• Print or export reports if required

6. SYSTEM SETTINGS MODULE

This module allows configuration and maintenance of the system.

• Manage system settings
• Control application features
• Maintain overall system performance

Conclusion

The modular structure of the **RailPlot System** improves system efficiency, maintainability, and scalability. Each module performs a specific function, making the system well-organized and easy to manage.

DATA STRUCTURE SNAP SHOT

The **Data Dictionary** is a centralized repository that stores the description of data elements, data structures, and system processes used in the application. It provides detailed information about each data item, including its name, type, and purpose.

The data dictionary plays an important role in database design, as it ensures consistency, accuracy, and proper organization of data within the system.

In the **RailPlot System**, the data dictionary is used to define and manage all database tables and fields. Proper care has been taken during the design phase to ensure that the database structure supports future expansion and additional system requirements.

USER DETAILS (Login)

Field Name	Data Type	Description
user_id	INT (PK)	Unique ID of user
username	VARCHAR	Username for login
password	VARCHAR	Encrypted password
role	VARCHAR	User role (Admin/User)
email	VARCHAR	Email address

RAIL DATA DETAILS (rail_data)

Field Name	Data Type	Description
data_id	INT (PK)	Unique data ID
title	VARCHAR	Data title
description	TEXT	Description of data
created_by	INT (FK)	User who created data
created_date	DATE	Date of creation

ROUTE DETAILS (route_details)

Field Name	Data Type	Description
route_id	INT (PK)	Unique route ID
source	VARCHAR	Starting location
destination	VARCHAR	Ending location

Field Name	Data Type	Description
distance	FLOAT	Distance between stations
status	VARCHAR	Active/Inactive

API LOG DETAILS (api_logs)

Field Name	Data Type	Description
log_id	INT (PK)	Unique log ID
api_name	VARCHAR	API name
request_time	DATETIME	Request time
response_status	VARCHAR	Status of response

REPORT DETAILS (reports)

Field Name	Data Type	Description
report_id	INT (PK)	Unique report ID
report_type	VARCHAR	Type of report
generated_by	INT (FK)	Generated by user
generated_date	DATE	Date of generation

SYSTEM SETTINGS (settings)

Field Name	DataType	Description
setting_id	INT (PK)	Unique setting ID
setting_name	VARCHAR	Name of setting
setting_value	VARCHAR	Value of setting

TRAIN DETAILS (train_details)

Field Name	Data Type	Description
train_id	INT (PK)	Unique train ID
train_name	VARCHAR	Name of the train
train_type	VARCHAR	Express / Passenger
total_seats	INT	Total seats available
route_id	INT (FK)	Associated route

STATION DETAILS (station_details)

Field Name **DataType Description**

Field Name	Data Type	Description
station_id	INT (PK)	Unique station ID
station_name	VARCHAR	Name of station
location	VARCHAR	City/location
code	VARCHAR	Station code

BOOKING DETAILS (booking_details)

Field Name	Data Type	Description
booking_id	INT (PK)	Unique booking ID
user_id	INT (FK)	User who booked
train_id	INT (FK)	Selected train
journey_date	DATE	Date of journey
seats_booked	INT	Number of seats
status	VARCHAR	Confirmed / Pending

PAYMENT DETAILS (payment_details)

Field Name	Data Type	Description
payment_id	INT (PK)	Unique payment

Field Name	DataType	Description
		ID
booking_id	INT (FK)	Booking reference
amount	FLOAT	Payment amount
payment_mode	VARCHAR	UPI / Card / Net Banking
payment_status	VARCHAR	Success / Failed
payment_date	DATE	Payment date

TICKET DETAILS (ticket_details)

Field Name	Data Type	Description
ticket_id	INT (PK)	Unique ticket ID
booking_id	INT (FK)	Booking reference
seat_number.	VARCHAR	Seat number
coach	VARCHAR	Coach number
issue_date	DATE	Ticket issue date

Field Name **DataType Description**

FEEDBACK DETAILS (feedback)

Field Name	Data Type	Description
feedback_id	INT (PK)	Unique feedback ID
user_id	INT (FK)	User who gave feedback
message	TEXT	Feedback message
rating	INT	Rating (1–5)
date	DATE	Feedback date

ADMIN DETAILS (admin)

Field Name	Data Type	Description
admin_id	INT (PK)	Unique admin ID
name	VARCHAR	Admin name
email	VARCHAR	Email address
password	VARCHAR	Login password

NOTIFICATION DETAILS

Field Name **DataType Description**

(notifications)

Field Name	Data Type	Description
notification_id	INT (PK)	Unique ID
user_id	INT (FK)	Target user
message	TEXT	Notification message
status	VARCHAR	Read / Unread
date	DATE	Notification date

LIST OF REPORTS

The following reports will be generated in the **RailPlot System** to help users analyze and manage data efficiently:

1. User Registration Report

This report contains the details of all registered users in the system. It includes information such as username, role, and registration details.

SCOPE AND FUTURE ENHANCEMENT

There is always scope for improvement in any system, and the **RailPlot System** is designed in a flexible and scalable manner to allow future enhancements. The current system fulfills the

basic requirements, but additional features and improvements can be implemented as needed.

The project is upgradeable, and modifications can be made according to user requirements. It can also be integrated as a sub-module in larger systems in the future.

Future Enhancements

Ability to **store and manage large volumes of railway-related data** efficiently

Further **reduction of manual effort** through automation of processes

Improved **processing speed**, reducing response time and delays

Addition of **advanced reporting and analytics features**

Integration with other systems or external APIs for enhanced functionality

Implementation of **advanced security features** for better data protection

Development of a **mobile-friendly interface or application**

IMPLEMENTATION METHODOLOGY

Implementation is the process of deploying the developed and tested system in a real working environment using actual hardware, software, and user interaction. It ensures that the system works correctly and efficiently in practical conditions.

For the **RailPlot System**, implementation is carried out using a web-based environment, which allows the system to run over the internet or intranet.

TESTING

Software testing is a critical phase of software development that ensures the system is free from errors and performs according to the specified requirements. It improves the reliability, performance, and quality of the software.

The testing process for the **RailPlot System** consists of the following stages:

a) TEST PLAN DEVELOPMENT

Test planning defines the scope, objectives, and approach for testing the system. It ensures that testing is conducted systematically with minimum time and effort.

The main objectives of testing are:
• To identify and remove errors
• To verify system functionality
• To ensure reliability and performance

b) TESTING STRATEGY

The testing strategy defines how testing will be carried out. It ensures flexibility and adaptability during the development process.

• Focus on both functional and non-functional testing
• Use of different testing techniques to detect errors
• Ensures complete coverage of all modules

c) UNIT TESTING

Unit testing focuses on testing individual modules of the system independently.

In the **RailPlot System**:
• Each module (User, Data Management, API, etc.) is tested separately
• Valid and invalid inputs are provided to check correctness
• Errors within modules are identified and fixed

Example:
• Login module is tested with correct and incorrect username/password
• System displays error message like **“Invalid username or password”** for wrong inputs

d) INTEGRATION TESTING

Integration testing is performed after unit testing, where all modules are combined and tested as a complete system.

• Ensures proper interaction between modules
• Checks data flow between frontend, backend, and database
• Identifies errors during module integration

e) VALIDATION TESTING

Validation testing ensures that the system meets all user requirements and performs as expected.

• System is tested using real-time scenarios
• Both normal and boundary inputs are used
• Ensures system reliability and accuracy

TEST CASE DESIGN

Test case design techniques are used to create effective test cases for identifying errors.

1. CONDITION TESTING

Condition testing verifies logical conditions in the program.

- Tests Boolean conditions and decision-making logic
- Ensures correct output for different conditions

2. BOUNDARY VALUE ANALYSIS

This technique focuses on testing boundary values of input data.

- Errors are more likely to occur at extreme values
- System is tested with minimum, maximum, and beyond-limit inputs

Example:
- Testing input fields with very large or invalid data
- System should display proper validation messages

3. EQUIVALENCE PARTITIONING

This is a black-box testing technique that divides input data into valid and invalid groups (classes).

• Reduces number of test cases
• Ensures coverage of all input types

Example:
• Valid input class → correct data format
• Invalid input class → incorrect or missing data

ENHANCEMENTS

Despite its advantages, the **RailPlot System** has certain limitations that may affect its performance and usability:

• The system depends on a **stable internet connection** for proper functioning
• Limited features in the current version compared to large-scale systems
• Data handling capacity may be restricted for extremely large datasets
• Security features are basic and may require further enhancement
• Performance may decrease with increased number of users or requests

---

RECOMMENDED AREAS FOR FURTHER RESEARCH

Considering future advancements and changing requirements, the following areas can be explored to improve the system:

1. Advanced Data Management

There is scope for implementing advanced techniques for handling large-scale data, such as:

- Efficient data storage and retrieval methods
- Data optimization and performance tuning

2. Automation and Smart Features

Future research can focus on:

- Automating data processing and updates
- Implementing intelligent features for better decision-making

3. System Integration

- Integration with external systems and APIs
- Real-time data synchronization

4. Security Enhancements

- Implementation of advanced authentication methods
- Data encryption and secure communication protocols

5. User Experience Improvement

• Developing a mobile application version
• Enhancing UI/UX for better usability

SECURITY MECHANISM

Implementation of security is a critical aspect of any software system. Different applications use different security mechanisms based on their requirements. In the **RailPlot System**, security is implemented at both user level and data level to ensure safe and controlled access.

User Authentication

The system authenticates users using a **username and password**. Only authorized users are allowed to access the system.

• Each user must log in using valid credentials
• Unauthorized access is restricted

• Invalid login attempts display an error message such as **“Invalid username or password”**

Role-Based Access Control

• The **Administrator** has full control over the system
• Admin can create, update, or delete user accounts
• Admin can grant or revoke permissions to users
• Normal users have limited access based on their role

Data-Level Security

Data security is maintained by applying validation rules and constraints at the database level.

• Only valid and verified data can be entered
• Prevents incorrect or duplicate data entries
• Ensures data integrity and consistency

Password Security

• Passwords are stored securely (preferably encrypted)
• Users can change their passwords when required
• Strong password practices are encouraged (combination of letters, numbers, etc.)

System Protection

- All sensitive information is protected through authentication
- Access to system data is restricted to authorized users only
- Secure communication between frontend and backend

SNAPSHOTS

HOME PAGE :

CODE :

```
import React, { useState, useEffect } from "react";
import { Link } from "react-router-dom";
```

```
import banner from "../../src/images/bannerRailwayN.png";
import sideBanner from
"../../src/images/railwaySideBanner.png";
import sideBanner2 from
"../../src/images/railwaySideBanner2.png";
import DemoRequestForm from
"../components/DemoRequestForm";
import { getDataFromLocalStorage } from
"../utils/localStorage";

const HomeComponent = () => {
  const [showDemoForm, setShowDemoForm] =
useState(false);
  const [userInfo, setUserInfo] = useState(null);
  useEffect(() => {
    const user = getDataFromLocalStorage("userInfo");
    setUserInfo(user);
  }, []);

  return (
    <div>
      <div className="w-full flex items-center ">
        <div className="bannerTitle w-full block max-w-
[1080px] mx-auto  p-[50px] pt-[0px] text-white absolute z-
[11]">
          <h1 className="text-[56px] pb-[15px] text-[#fff]">
            Powerful Data Analytics for Railways
          </h1>
```

```
        <p className="max-w-[675px] text-[17px] mb-[15px]">
          Analyze, improve, and guide loco pilots with real-time
insights for
          safer, more efficient railway operations across India.
        </p>
        {!userInfo && (
          <div>
            <span
              onClick={() => setShowDemoForm(true)}
              id="getStartedButton"
              className=" cursor-pointer reportGenerateBg
rounded-[4px] text-[#fff] sm:text-[16px] text-[14px] sm:px-
[15px] px-[12px] py-[8px] sm:py-[15px] inline-block mt-[20px]"
            >
              REQUEST A DEMO
            </span>
          </div>
        )}
      </div>
      <div className="banner">
        <img src={banner} className="w-full" />
      </div>
    </div>
    <div className="w-full">
      <div className="max-w-[1200px] w-full mx-auto sm:my-
[80px] my-[40px] px-4">
        <div className="w-full powerfulDataBox">
          {/* <h3 className="text-[34px] font-semibold">
```

```
            Powerful Data Analytics for Railways
          </h3> */}
          <div className="w-full flex flex-row
speedTimeGraph">
            <div className="w-[60%] pr-[20px]">
              <ul>
                <li>
                  <h4 className="text-[24px] text-[#9b4b90]">
                    Speed-Time Graphs
                  </h4>
                  <p>
                    Visualize loco speed trends to monitor
operational
                    efficiency and punctuality with precise real-time
data
                    tracking.
                  </p>
                </li>
                <li>
                  <h4 className="text-[24px] text-[#9b4b90]">
                    Braking Profile Analysis
                  </h4>
                  <p>
                    Analyze braking patterns to enhance safety and
optimize
                    braking efficiency, reducing wear and increasing
train
                    stability.
```

```
                </p>
              </li>
              <li>
                <h4 className="text-[24px] text-[#9b4b90]">
                  Guideline Violation Detection
                </h4>
                <p>
                  Detect overspeed, late braking, and other violations
                  instantly to enforce compliance and prevent accidents.
                </p>
              </li>
            </ul>
          </div>
          <div className="w-[40%] speedTimeBanner pl-[20px]">
            <img src={sideBanner} className="w-full" />
          </div>
        </div>
      </div>
    </div>
    <div className="w-full reportGenerateBg">
      <div className="max-w-[1200px] w-full mx-auto sm:my-[80px] my-[40px] py-16 px-4">
        <div className="bg-white sm:my-4 my-0 p-2 rounded-[10px] insightFlSpeedCol">
```

```
          <h3 className="text-[30px] text-center py-4 text-
[#9b4b90] font-semibold">
            Insightful Speed-Time Curve Comparison
          </h3>
          <div className="w-full">
            <ul className="px-2 flex flex-row justify-between
max-w-[980px] mx-auto my-8">
              <li className="flex flex-row mr-2 w-[40%] mb-4">
                <span className="w-[32px] mr-2">
                  <svg
                    className="colorable-icon w-[32px] h-[32px]
mt-2 mr-4"
                    viewBox="0 0 512 512"
                    xmlns="http://www.w3.org/2000/svg"
                    data-icon="chart-line-up-down"
                    data-prefix="fal"
                    aria-hidden="true"
                  >
                    <path
                      d="M32 48c0-8.8-7.2-16-16-16S0 39.2 0
48V400c0 44.2 35.8 80 80 80H496c8.8 0 16-7.2 16-16s-7.2-
16-16-16H80c-26.5 0-48-21.5-48-48V48zM368 96h57.4L288
233.4l-68.7-68.7c-3-3-7.1-4.7-11.3-4.7s-8.3 1.7-11.3 4.7l-96
96c-6.2 6.2-6.2 16.4 0 22.6s16.4 6.2 22.6 0L208 198.6l68.7
68.7c6.2 6.2 16.4 6.2 22.6 0L448 118.6V176c0 8.8 7.2 16 16
16s16-7.2 16-16V80c0-8.8-7.2-16-16-16H368c-8.8 0-16 7.2-
16 16s7.2 16 16 16zm0 288h96c8.8 0 16-7.2 16-16V272c0-
8.8-7.2-16-16-16s-16 7.2-16 16v57.4l-68.7-68.7-22.6
```

```
22.6L425.4 352H368c-8.8 0-16 7.2-16 16s7.2 16 16 16z"
                  fill="currentColor"
                ></path>
              </svg>
            </span>
            <div>
              <h4 className="text-[22px] font-semibold">
                Before Training
              </h4>
              <p>
                Irregular speed, sharp accelerations and braking
reducing
                efficiency.
              </p>
            </div>
          </li>
          <li className="flex flex-row mr-2 w-[40%]  mb-4">
            <span className="w-[32px] mr-2">
              <svg
                className="colorable-icon w-[32px] h-[32px]
mt-2 mr-4"
                viewBox="0 0 512 512"
                xmlns="http://www.w3.org/2000/svg"
                data-icon="chart-line-up"
                data-prefix="fal"
                aria-hidden="true"
              >
                <path
```

```
                    d="M32 48c0-8.8-7.2-16-16-16S0 39.2 0
48V400c0 44.2 35.8 80 80 80H496c8.8 0 16-7.2 16-16s-7.2-
16-16-16H80c-26.5 0-48-21.5-48-48V48zm288 96c0 8.8 7.2
16 16 16h89.4L320 265.4l-84.7-84.7c-6.2-6.2-16.4-6.2-22.6
0l-112 112c-6.2 6.2-6.2 16.4 0 22.6s16.4 6.2 22.6 0L224
214.6l84.7 84.7c6.2 6.2 16.4 6.2 22.6 0L448 182.6V272c0 8.8
7.2 16 16 16s16-7.2 16-16V144c0-8.8-7.2-16-16-16H336c-8.8
0-16 7.2-16 16z"
                    fill="currentColor"
                  ></path>
                </svg>
              </span>
              <div>
                <h4 className="text-[22px] font-semibold">
                  After Training
                </h4>
                <p>
                  Smoother driving with optimal speed control
enhancing
                  safety and fuel economy.
                </p>
              </div>
            </li>
          </ul>
        </div>
      </div>
    </div>
  </div>
```

```
<div className="w-full">
  <div className="max-w-[1200px] w-full mx-auto sm:my-[80px] my-[40px] px-4">
    <div className="w-full powerfulDataBox">
      <div className="w-full flex flex-row breakzoneBox">
        <div className="w-[40%] pl-[20px]">
          <img src={sideBanner2} className="w-full" />
        </div>
        <div className="w-[60%] pl-[20px]">
          <h3 className="text-[34px] font-semibold text-[#9b4b90]">
            Braking Zone Visual Analysis
          </h3>
          <ul>
            <li>
              <p>
                This visual tool identifies critical braking zones,
                flagging late applications and unnecessary early brakes
                for targeted pilot training.
              </p>
            </li>
            <li>
              <p>
                Reducing braking inconsistencies ensures greater passenger
                comfort, less mechanical stress, and enhanced safety.
```

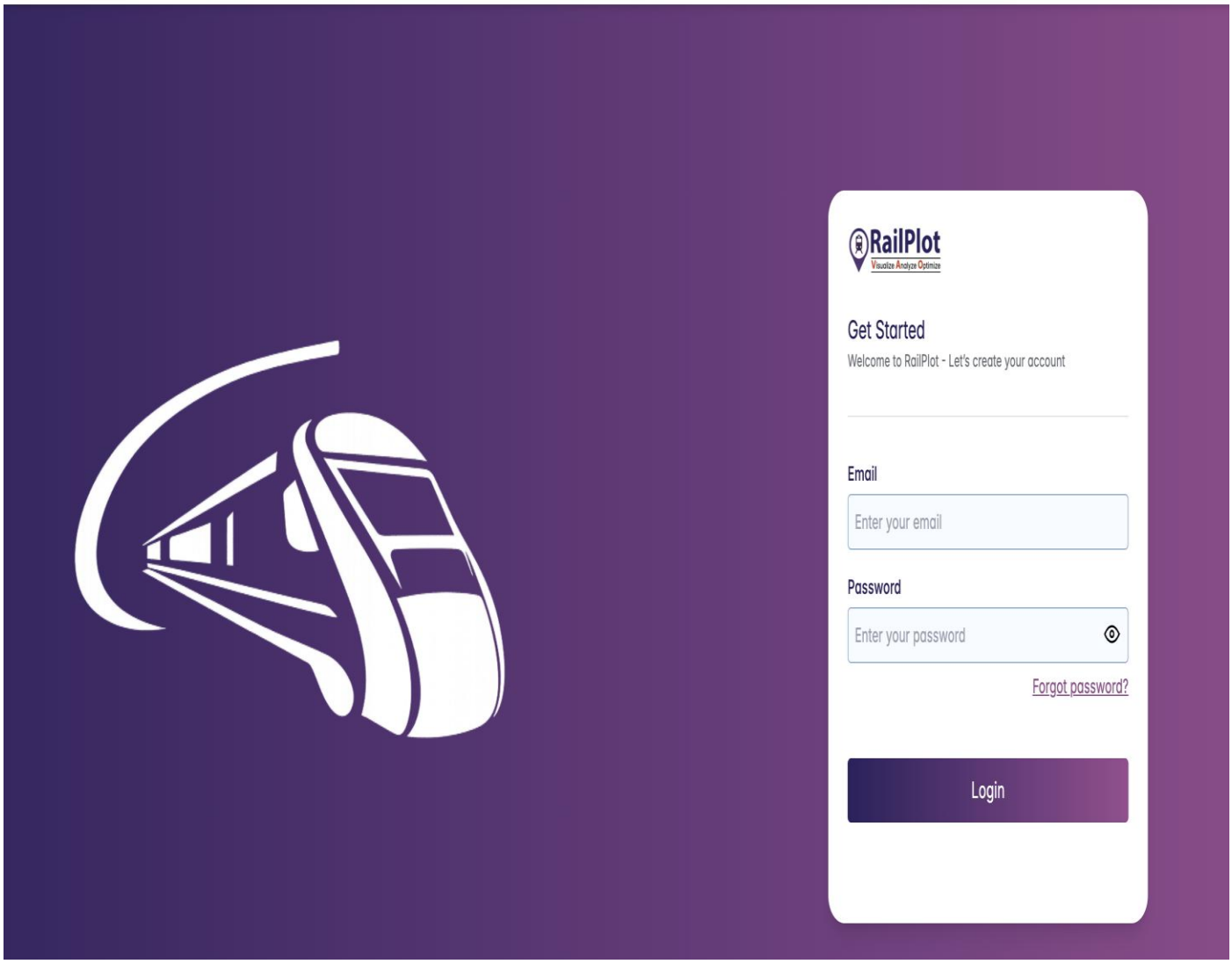
```
                </p>
              </li>
            </ul>
          </div>
        </div>
      </div>
    </div>
    <div className="w-full bg-[#efefef]">
      <div className="max-w-[1200px] w-full mx-auto sm:py-
[80px] py-[40px] px-4">
        <div className="w-full p-2 drivingPatternBox">
          <h3 className="text-center text-[26px] mb-8 font-
semibold">
            Driving Pattern Suggestions
          </h3>
          <ul className="flex flex-row justify-bweteen">
            <li className=" bg-white p-4  rounded-[5px]">
              <span>
                <h4 className="text-[20px] font-semibold">
                  Optimized Speed Control
                </h4>
                <p className="text-[#434343]">
                  Encourages steady acceleration to reduce energy
consumption.
                </p>
              </span>
            </li>
```

```
            <li className=" bg-white p-4 mx-4 rounded-[5px]">
              <span>
                <h4 className="text-[20px] font-semibold">Timely
Braking</h4>
                <p className="text-[#434343]">
                  Promotes early and smooth braking for increased
safety and
                  efficiency.
                </p>
              </span>
            </li>
            <li className=" bg-white p-4  rounded-[5px]">
              <span>
                <h4 className="text-[20px] font-semibold">
                  Compliance Reinforcement
                </h4>
                <p className="text-[#434343]">
                  Alerts on any deviations to ensure adherence to
safety
                  guidelines.
                </p>
              </span>
            </li>
          </ul>
        </div>
      </div>
    </div>
    {showDemoForm && (
```

```
        <DemoRequestForm onClose={() =>
setShowDemoForm(false)} />
      )}
    </div>
  );
};

export default HomeComponent;
```

LOGIN PAGE :

RailPlot
Visualize Analyze Optimize



CODE : import React, { useState, useEffect } from "react";
import { Link } from "react-router-dom";

```
import banner from "../../src/images/bannerRailwayN.png";
import sideBanner from
"../../src/images/railwaySideBanner.png";
import sideBanner2 from
"../../src/images/railwaySideBanner2.png";
import DemoRequestForm from
"../components/DemoRequestForm";
import { getDataFromLocalStorage } from
"../utils/localStorage";

const HomeComponent = () => {
  const [showDemoForm, setShowDemoForm] =
useState(false);
  const [userInfo, setUserInfo] = useState(null);
  useEffect(() => {
    const user = getDataFromLocalStorage("userInfo");
    setUserInfo(user);
  }, []);

  return (
    <div>
      <div className="w-full flex items-center ">
        <div className="bannerTitle w-full block max-w-
[1080px] mx-auto  p-[50px] pt-[0px] text-white absolute z-
[11]">
          <h1 className="text-[56px] pb-[15px] text-[#fff]">
            Powerful Data Analytics for Railways
          </h1>
```

```
        <p className="max-w-[675px] text-[17px] mb-[15px]">
          Analyze, improve, and guide loco pilots with real-time
insights for
          safer, more efficient railway operations across India.
        </p>
        {!userInfo && (
          <div>
            <span
              onClick={() => setShowDemoForm(true)}
              id="getStartedButton"
              className=" cursor-pointer reportGenerateBg
rounded-[4px] text-[#fff] sm:text-[16px] text-[14px] sm:px-
[15px] px-[12px] py-[8px] sm:py-[15px] inline-block mt-[20px]"
            >
              REQUEST A DEMO
            </span>
          </div>
        )}
      </div>
      <div className="banner">
        <img src={banner} className="w-full" />
      </div>
    </div>
    <div className="w-full">
      <div className="max-w-[1200px] w-full mx-auto sm:my-
[80px] my-[40px] px-4">
        <div className="w-full powerfulDataBox">
          {/* <h3 className="text-[34px] font-semibold">
```

```
            Powerful Data Analytics for Railways
          </h3> */}
          <div className="w-full flex flex-row
speedTimeGraph">
            <div className="w-[60%] pr-[20px]">
              <ul>
                <li>
                  <h4 className="text-[24px] text-[#9b4b90]">
                    Speed-Time Graphs
                  </h4>
                  <p>
                    Visualize loco speed trends to monitor
operational
                    efficiency and punctuality with precise real-time
data
                    tracking.
                  </p>
                </li>
                <li>
                  <h4 className="text-[24px] text-[#9b4b90]">
                    Braking Profile Analysis
                  </h4>
                  <p>
                    Analyze braking patterns to enhance safety and
optimize
                    braking efficiency, reducing wear and increasing
train
                    stability.
```

```
                </p>
              </li>
              <li>
                <h4 className="text-[24px] text-[#9b4b90]">
                  Guideline Violation Detection
                </h4>
                <p>
                  Detect overspeed, late braking, and other violations
                  instantly to enforce compliance and prevent accidents.
                </p>
              </li>
            </ul>
          </div>
          <div className="w-[40%] speedTimeBanner pl-[20px]">
            <img src={sideBanner} className="w-full" />
          </div>
        </div>
      </div>
    </div>
    <div className="w-full reportGenerateBg">
      <div className="max-w-[1200px] w-full mx-auto sm:my-[80px] my-[40px] py-16 px-4">
        <div className="bg-white sm:my-4 my-0 p-2 rounded-[10px] insightFlSpeedCol">
```

```
          <h3 className="text-[30px] text-center py-4 text-
[#9b4b90] font-semibold">
            Insightful Speed-Time Curve Comparison
          </h3>
          <div className="w-full">
            <ul className="px-2 flex flex-row justify-between
max-w-[980px] mx-auto my-8">
              <li className="flex flex-row mr-2 w-[40%] mb-4">
                <span className="w-[32px] mr-2">
                  <svg
                    className="colorable-icon w-[32px] h-[32px]
mt-2 mr-4"
                    viewBox="0 0 512 512"
                    xmlns="http://www.w3.org/2000/svg"
                    data-icon="chart-line-up-down"
                    data-prefix="fal"
                    aria-hidden="true"
                  >
                    <path
                      d="M32 48c0-8.8-7.2-16-16-16S0 39.2 0
48V400c0 44.2 35.8 80 80 80H496c8.8 0 16-7.2 16-16s-7.2-
16-16-16H80c-26.5 0-48-21.5-48-48V48zM368 96h57.4L288
233.4l-68.7-68.7c-3-3-7.1-4.7-11.3-4.7s-8.3 1.7-11.3 4.7l-96
96c-6.2 6.2-6.2 16.4 0 22.6s16.4 6.2 22.6 0L208 198.6l68.7
68.7c6.2 6.2 16.4 6.2 22.6 0L448 118.6V176c0 8.8 7.2 16 16
16s16-7.2 16-16V80c0-8.8-7.2-16-16-16H368c-8.8 0-16 7.2-
16 16s7.2 16 16 16zm0 288h96c8.8 0 16-7.2 16-16V272c0-
8.8-7.2-16-16-16s-16 7.2-16 16v57.4l-68.7-68.7-22.6
```

```
22.6L425.4 352H368c-8.8 0-16 7.2-16 16s7.2 16 16 16z"
                  fill="currentColor"
                ></path>
              </svg>
            </span>
            <div>
              <h4 className="text-[22px] font-semibold">
                Before Training
              </h4>
              <p>
                Irregular speed, sharp accelerations and braking
reducing
                efficiency.
              </p>
            </div>
          </li>
          <li className="flex flex-row mr-2 w-[40%]  mb-4">
            <span className="w-[32px] mr-2">
              <svg
                className="colorable-icon w-[32px] h-[32px]
mt-2 mr-4"
                viewBox="0 0 512 512"
                xmlns="http://www.w3.org/2000/svg"
                data-icon="chart-line-up"
                data-prefix="fal"
                aria-hidden="true"
              >
                <path
```

```
                    d="M32 48c0-8.8-7.2-16-16-16S0 39.2 0
48V400c0 44.2 35.8 80 80 80H496c8.8 0 16-7.2 16-16s-7.2-
16-16-16H80c-26.5 0-48-21.5-48-48V48zm288 96c0 8.8 7.2
16 16 16h89.4L320 265.4l-84.7-84.7c-6.2-6.2-16.4-6.2-22.6
0l-112 112c-6.2 6.2-6.2 16.4 0 22.6s16.4 6.2 22.6 0L224
214.6l84.7 84.7c6.2 6.2 16.4 6.2 22.6 0L448 182.6V272c0 8.8
7.2 16 16 16s16-7.2 16-16V144c0-8.8-7.2-16-16-16H336c-8.8
0-16 7.2-16 16z"
                    fill="currentColor"
                  ></path>
                </svg>
              </span>
              <div>
                <h4 className="text-[22px] font-semibold">
                  After Training
                </h4>
                <p>
                  Smoother driving with optimal speed control
enhancing
                  safety and fuel economy.
                </p>
              </div>
            </li>
          </ul>
        </div>
      </div>
    </div>
  </div>
```

```
<div className="w-full">
  <div className="max-w-[1200px] w-full mx-auto sm:my-[80px] my-[40px] px-4">
    <div className="w-full powerfulDataBox">
      <div className="w-full flex flex-row breakzoneBox">
        <div className="w-[40%] pl-[20px]">
          <img src={sideBanner2} className="w-full" />
        </div>
        <div className="w-[60%] pl-[20px]">
          <h3 className="text-[34px] font-semibold text-[#9b4b90]">
            Braking Zone Visual Analysis
          </h3>
          <ul>
            <li>
              <p>
                This visual tool identifies critical braking zones,
                flagging late applications and unnecessary early brakes
                for targeted pilot training.
              </p>
            </li>
            <li>
              <p>
                Reducing braking inconsistencies ensures greater passenger
                comfort, less mechanical stress, and enhanced safety.
```

```
                </p>
              </li>
            </ul>
          </div>
        </div>
      </div>
    </div>
    <div className="w-full bg-[#efefef]">
      <div className="max-w-[1200px] w-full mx-auto sm:py-
[80px] py-[40px] px-4">
        <div className="w-full p-2 drivingPatternBox">
          <h3 className="text-center text-[26px] mb-8 font-
semibold">
            Driving Pattern Suggestions
          </h3>
          <ul className="flex flex-row justify-bweteen">
            <li className=" bg-white p-4  rounded-[5px]">
              <span>
                <h4 className="text-[20px] font-semibold">
                  Optimized Speed Control
                </h4>
                <p className="text-[#434343]">
                  Encourages steady acceleration to reduce energy
consumption.
                </p>
              </span>
            </li>
```

```
            <li className=" bg-white p-4 mx-4 rounded-[5px]">
              <span>
                <h4 className="text-[20px] font-semibold">Timely
Braking</h4>
                <p className="text-[#434343]">
                  Promotes early and smooth braking for increased
safety and
                  efficiency.
                </p>
              </span>
            </li>
            <li className=" bg-white p-4  rounded-[5px]">
              <span>
                <h4 className="text-[20px] font-semibold">
                  Compliance Reinforcement
                </h4>
                <p className="text-[#434343]">
                  Alerts on any deviations to ensure adherence to
safety
                  guidelines.
                </p>
              </span>
            </li>
          </ul>
        </div>
      </div>
    </div>
    {showDemoForm && (
```

```
        <DemoRequestForm onClose={() =>
setShowDemoForm(false)} />
      )}
    </div>
  );
};

export default HomeComponent;
```

DASHBOARD :

RailPlot
Visualize Analyze Optimize

Home Dashboard Control Center

Dashboard

+ Analyze New

Date Range : 13/02/2026 - 13/02/2026 LP CMS : Filter Clear Download All PDF

18140	BWN-ASN[GPS-Data] 13/02/2026
Analyzed By :	ARBIND KUMAR
Route:	[19] UDL → BQT
LP ID :	UDL2719
LP Name :	RANJAN RABIDAS R...
Departure Time :	10/02/2026 15:20:14
Arrival Time :	10/02/2026 18:55:27
Average Speed :	1.35
Total Distance (KM) :	4.846
No Violations ✓	
Delete Source File View Download Whatsapp	

18139	ASN-BWN[GPS-Data] 12/02/2026
Analyzed By :	SUDIP SAHA
Route:	[18] POL → BWN
LP ID :	UDL4461
LP Name :	AJIT KUMAR II
Departure Time :	02/11/2026 15:15:54
Arrival Time :	02/11/2026 23:17:06
Average Speed :	9.45
Total Distance (KM) :	75.7919
No Violations ✓	
Delete Source File View Download Whatsapp	

18137	SNT-UDL[GPS-Data] 12/02/2026
Analyzed By :	SUDIP SAHA
Route:	[25] PCR → UDL
LP ID :	UDL4461
LP Name :	AJIT KUMAR II
Departure Time :	02/08/2026 09:54:33
Arrival Time :	02/08/2026 16:47:24
Average Speed :	4.28
Total Distance (KM) :	29.4484
No Violations ✓	
Delete Source File View Download Whatsapp	

18129	SNT-UDL[GPS-Data] 12/02/2026
Analyzed By :	SUDIP SAHA
Route:	[25] SNT → DUJ
LP ID :	PAW1028
LP Name :	UPENDRA KUMAR L...
Departure Time :	02/11/2026 21:20:18
Arrival Time :	02/11/2026 23:59:58
Average Speed :	15.06
Total Distance (KM) :	40.0703
No Violations ✓	
Delete Source File View Download Whatsapp	

18124	ASN-BWN[GPS-Data] 12/02/2026
Analyzed By :	RANJEET PASWAN
Route:	[18] UDL → GLI
LP ID :	UDL2276
LP Name :	PINTU KUMAR IV LP...
Departure Time :	02/10/2026 03:05:32

18123	SNT-UDL[GPS-Data] 12/02/2026
Analyzed By :	Amit Anand
Route:	[25] SNT → DUJ
LP ID :	PAW1213
LP Name :	AJIT KUMAR IV DUAL
Departure Time :	02/08/2026 14:15:47

18122	BWN-GMO[GPS-Data] 12/02/2026
Analyzed By :	PAW LOBBY
Route:	[29] KAN → BRR
LP ID :	
LP Name :	
Departure Time :	02/11/2026 00:43:17

18121	GMO-ASN[GPS-Data] 12/02/2026
Analyzed By :	PAW LOBBY
Route:	[23] KMME → STN
LP ID :	
LP Name :	
Departure Time :	02/11/2026 04:20:56

RailPlot
Visualize Analyze Optimize

Home Dashboard Control Center

Templates

+ Add Template

[2] ASN-GMO UP
Attacking Speed File:
Gradient File:
ISD File:
PSR File:
Station File:
Delete Use It Details Edit

[3] ASN-JAJ UP
Attacking Speed File:
Gradient File:
ISD File:
PSR File:
Station File:
Delete Use It Details Edit

[4] BWN-ASN UP
ISD File:
PSR File:
Station File:
Delete Use It Details Edit

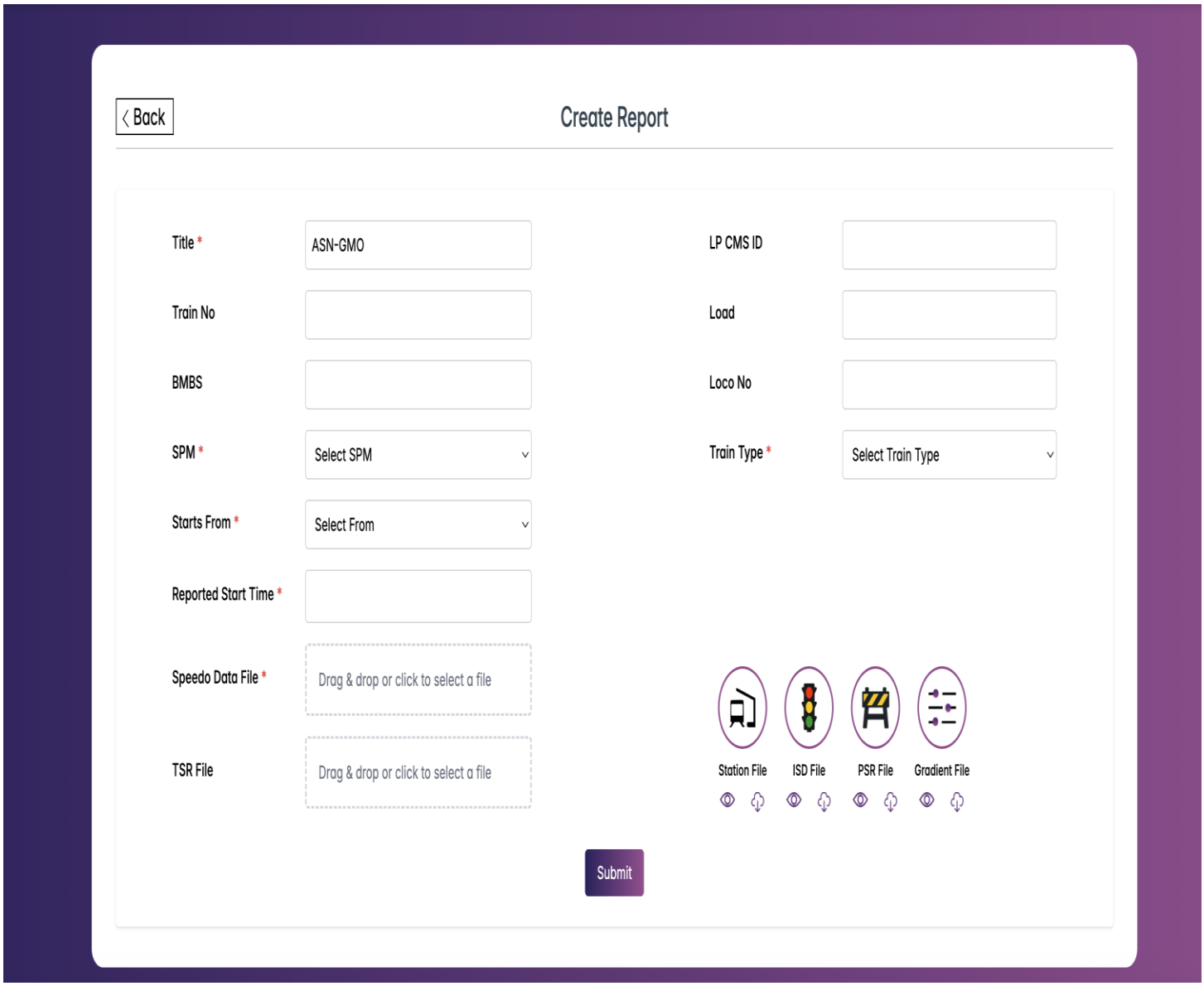
[8] BWN-GMO UP
Attacking Speed File:
Gradient File:
ISD File:
PSR File:
Station File:
Delete Use It Details Edit

[9] BWN-JAJ UP
Attacking Speed File:
Gradient File:
ISD File:
PSR File:
Station File:
Delete Use It Details Edit

[10] UDL-SNT UP
Gradient File:
ISD File:
Station File:
Delete Use It Details Edit

[19] BWN-ASN[GPS-Data] UP
ISD File:
PSR File:
Station File:
Delete Use It Details Edit

[20] ASN-JAJ[GPS-Data] UP
Attacking Speed File:
Gradient File:
ISD File:
PSR File:
Station File:
Delete Use It Details Edit


RailPlot
Visualize Analyze Optimize
Home
Dashboard
Control Center
Back
Create Report
Title *
ASN-GMO
LP CMS ID
Train No
Load
BMBS
Loco No
SPM *
Select SPM
Train Type *
Select Train Type
Starts From *
Select From
Reported Start Time *
Speedo Data File *
Drag & drop or click to select a file
TSR File
Drag & drop or click to select a file
Station File
ISD File
PSR File
Gradient File
Submit

Code :

```
import React, { useState, useEffect } from "react";
import { Link } from "react-router-dom";
import banner from "../../src/images/bannerRailwayN.png";
import sideBanner from
"../../src/images/railwaySideBanner.png";
import sideBanner2 from
"../../src/images/railwaySideBanner2.png";
import DemoRequestForm from
"../components/DemoRequestForm";
import { getDataFromLocalStorage } from
"../utils/localStorage";

const HomeComponent = () => {
  const [showDemoForm, setShowDemoForm] =
useState(false);
  const [userInfo, setUserInfo] = useState(null);
  useEffect(() => {
    const user = getDataFromLocalStorage("userInfo");
    setUserInfo(user);
  }, []);

  return (
    <div>
      <div className="w-full flex items-center ">
        <div className="bannerTitle w-full block max-w-
[1080px] mx-auto  p-[50px] pt-[0px] text-white absolute z-
```

```
[11]">
        <h1 className="text-[56px] pb-[15px] text-[#fff]">
          Powerful Data Analytics for Railways
        </h1>
        <p className="max-w-[675px] text-[17px] mb-[15px]">
          Analyze, improve, and guide loco pilots with real-time
insights for
          safer, more efficient railway operations across India.
        </p>
        {!userInfo && (
          <div>
            <span
              onClick={() => setShowDemoForm(true)}
              id="getStartedButton"
              className=" cursor-pointer reportGenerateBg
rounded-[4px] text-[#fff] sm:text-[16px] text-[14px] sm:px-
[15px] px-[12px] py-[8px] sm:py-[15px] inline-block mt-[20px]"
            >
              REQUEST A DEMO
            </span>
          </div>
        )}
      </div>
      <div className="banner">
        <img src={banner} className="w-full" />
      </div>
    </div>
    <div className="w-full">
```

```
      <div className="max-w-[1200px] w-full mx-auto sm:my-
[80px] my-[40px] px-4">
        <div className="w-full powerfulDataBox">
          {/* <h3 className="text-[34px] font-semibold">
            Powerful Data Analytics for Railways
          </h3> */}
          <div className="w-full flex flex-row
speedTimeGraph">
            <div className="w-[60%] pr-[20px]">
              <ul>
                <li>
                  <h4 className="text-[24px] text-[#9b4b90]">
                    Speed-Time Graphs
                  </h4>
                  <p>
                    Visualize loco speed trends to monitor
operational
                    efficiency and punctuality with precise real-time
data
                    tracking.
                  </p>
                </li>
                <li>
                  <h4 className="text-[24px] text-[#9b4b90]">
                    Braking Profile Analysis
                  </h4>
                  <p>
                    Analyze braking patterns to enhance safety and
```

```
optimize
                    braking efficiency, reducing wear and increasing
train
                    stability.
                  </p>
                </li>
                <li>
                  <h4 className="text-[24px] text-[#9b4b90]">
                    Guideline Violation Detection
                  </h4>
                  <p>
                    Detect overspeed, late braking, and other
violations
                    instantly to enforce compliance and prevent
accidents.
                  </p>
                </li>
              </ul>
            </div>
            <div className="w-[40%] speedTimeBanner pl-
[20px]">
              <img src={sideBanner} className="w-full" />
            </div>
          </div>
        </div>
      </div>
    </div>
    <div className="w-full reportGenerateBg">
```

```
      <div className="max-w-[1200px] w-full mx-auto sm:my-
[80px] my-[40px] py-16 px-4">
        <div className="bg-white sm:my-4 my-0 p-2 rounded-
[10px] insightFlSpeedCol">
          <h3 className="text-[30px] text-center py-4 text-
[#9b4b90] font-semibold">
            Insightful Speed-Time Curve Comparison
          </h3>
          <div className="w-full">
            <ul className="px-2 flex flex-row justify-between
max-w-[980px] mx-auto my-8">
              <li className="flex flex-row mr-2 w-[40%] mb-4">
                <span className="w-[32px] mr-2">
                  <svg
                    className="colorable-icon w-[32px] h-[32px]
mt-2 mr-4"
                    viewBox="0 0 512 512"
                    xmlns="http://www.w3.org/2000/svg"
                    data-icon="chart-line-up-down"
                    data-prefix="fal"
                    aria-hidden="true"
                  >
                    <path
                      d="M32 48c0-8.8-7.2-16-16-16S0 39.2 0
48V400c0 44.2 35.8 80 80 80H496c8.8 0 16-7.2 16-16s-7.2-
16-16-16H80c-26.5 0-48-21.5-48-48V48zM368 96h57.4L288
233.4l-68.7-68.7c-3-3-7.1-4.7-11.3-4.7s-8.3 1.7-11.3 4.7l-96
96c-6.2 6.2-6.2 16.4 0 22.6s16.4 6.2 22.6 0L208 198.6l68.7
```

```
68.7c6.2 6.2 16.4 6.2 22.6 0L448 118.6V176c0 8.8 7.2 16 16
16s16-7.2 16-16V80c0-8.8-7.2-16-16-16H368c-8.8 0-16 7.2-
16 16s7.2 16 16 16zm0 288h96c8.8 0 16-7.2 16-16V272c0-
8.8-7.2-16-16-16s-16 7.2-16 16v57.4l-68.7-68.7-22.6
22.6L425.4 352H368c-8.8 0-16 7.2-16 16s7.2 16 16 16z"
                  fill="currentColor"
                ></path>
              </svg>
            </span>
            <div>
              <h4 className="text-[22px] font-semibold">
                Before Training
              </h4>
              <p>
                Irregular speed, sharp accelerations and braking
reducing
                efficiency.
              </p>
            </div>
          </li>
          <li className="flex flex-row mr-2 w-[40%]  mb-4">
            <span className="w-[32px] mr-2">
              <svg
                className="colorable-icon w-[32px] h-[32px]
mt-2 mr-4"
                viewBox="0 0 512 512"
                xmlns="http://www.w3.org/2000/svg"
                data-icon="chart-line-up"
```

```
                data-prefix="fal"
                aria-hidden="true"
              >
                <path
                  d="M32 48c0-8.8-7.2-16-16-16S0 39.2 0
48V400c0 44.2 35.8 80 80 80H496c8.8 0 16-7.2 16-16s-7.2-
16-16-16H80c-26.5 0-48-21.5-48-48V48zm288 96c0 8.8 7.2
16 16 16h89.4L320 265.4l-84.7-84.7c-6.2-6.2-16.4-6.2-22.6
0l-112 112c-6.2 6.2-6.2 16.4 0 22.6s16.4 6.2 22.6 0L224
214.6l84.7 84.7c6.2 6.2 16.4 6.2 22.6 0L448 182.6V272c0 8.8
7.2 16 16 16s16-7.2 16-16V144c0-8.8-7.2-16-16-16H336c-8.8
0-16 7.2-16 16z"
                  fill="currentColor"
                ></path>
              </svg>
            </span>
            <div>
              <h4 className="text-[22px] font-semibold">
                After Training
              </h4>
              <p>
                Smoother driving with optimal speed control
enhancing
                safety and fuel economy.
              </p>
            </div>
          </li>
        </ul>
```

```
          </div>
        </div>
      </div>
    </div>
    <div className="w-full">
      <div className="max-w-[1200px] w-full mx-auto sm:my-
[80px] my-[40px] px-4">
        <div className="w-full powerfulDataBox">
          <div className="w-full flex flex-row breakzoneBox">
            <div className="w-[40%] pl-[20px]">
              <img src={sideBanner2} className="w-full" />
            </div>
            <div className="w-[60%] pl-[20px]">
              <h3 className="text-[34px] font-semibold text-
[#9b4b90]">
                Braking Zone Visual Analysis
              </h3>
              <ul>
                <li>
                  <p>
                    This visual tool identifies critical braking zones,
                    flagging late applications and unnecessary early
brakes
                    for targeted pilot training.
                  </p>
                </li>
                <li>
                  <p>
```

```
                    Reducing braking inconsistencies ensures
greater passenger
                    comfort, less mechanical stress, and enhanced
safety.
                  </p>
                </li>
              </ul>
            </div>
          </div>
        </div>
      </div>
      <div className="w-full bg-[#efefef]">
        <div className="max-w-[1200px] w-full mx-auto sm:py-
[80px] py-[40px] px-4">
          <div className="w-full p-2 drivingPatternBox">
            <h3 className="text-center text-[26px] mb-8 font-
semibold">
              Driving Pattern Suggestions
            </h3>
            <ul className="flex flex-row justify-bweteen">
              <li className=" bg-white p-4  rounded-[5px]">
                <span>
                  <h4 className="text-[20px] font-semibold">
                    Optimized Speed Control
                  </h4>
                  <p className="text-[#434343]">
                    Encourages steady acceleration to reduce energy
```

```
consumption.
                </p>
              </span>
            </li>
            <li className=" bg-white p-4 mx-4 rounded-[5px]">
              <span>
                <h4 className="text-[20px] font-semibold">Timely
Braking</h4>
                <p className="text-[#434343]">
                  Promotes early and smooth braking for increased
safety and
                  efficiency.
                </p>
              </span>
            </li>
            <li className=" bg-white p-4  rounded-[5px]">
              <span>
                <h4 className="text-[20px] font-semibold">
                  Compliance Reinforcement
                </h4>
                <p className="text-[#434343]">
                  Alerts on any deviations to ensure adherence to
safety
                  guidelines.
                </p>
              </span>
            </li>
          </ul>
```

```
          </div>
        </div>
      </div>
      {showDemoForm && (
        <DemoRequestForm onClose={() =>
setShowDemoForm(false)} />
      )}
    </div>
  );
};

export default HomeComponent;
```

CODE :

```
import React from "react";
import DashboardComponent from
"../components/DashboardComponent";

function Dashboard() {
  return (
    <div>
      <DashboardComponent />
    </div>
  );
```

```
}

export default Dashboard;


CODE : @tailwind base;
@tailwind components;
@tailwind utilities;

.reportGenerateBg {
  background-image: linear-gradient(270deg, #9b4b90 0,
#2c215d);
}

.no-print {
  display: none;
}

.loaderPdf {
  position: absolute;
  z-index: 77;
  left: 0%;
  top: 0px;
  display: flex;
  justify-content: center;
  align-items: center;
  width: 100%;
  height: 100%;
```

```
  background: rgba(0, 0, 0, 0.4);
}
.loaderPdf img {
  display: inline-block;
  width: 40px;
  animation: rotateAnimation 2s linear infinite;
}
@keyframes rotateAnimation {
  0% {
    transform: rotate(0deg);
  }

  to {
    transform: rotate(1turn);
  }
}
.logoHeader {
  display: flex;
  align-items: center;
  justify-content: center;
}
.logo {
  width: 350px;
}
.navIcon {
  display: none;
}
.dashboardCardUi li {
```

```
  display: flex;
  justify-content: space-between;
}
.loader {
  position: fixed;
  z-index: 77;
  left: 0%;
  top: 0%;
  display: flex;
  justify-content: center;
  align-items: center;
  width: 100%;
  height: 100%;
  background: rgba(0, 0, 0, 0.4);
}
.loader img {
  display: inline-block;
  width: 40px;
  animation: rotateAnimation 2s linear infinite;
}

.componentLoader {
  z-index: 77;
  left: 0%;
  top: 0%;
  display: flex;
  justify-content: center;
  align-items: center;
```

```
  width: 100%;
  height: 200px;
  background: #fff;
}

.componentLoader img {
  display: inline-block;
  width: 40px;
  animation: rotateAnimation 2s linear infinite;
}

body.loading {
  border: 5px solid red;
}
body.loading:before {
  content: "";
  width: 100%;
  height: 100%;
  background: rgba(0, 0, 0, 0.5);
  position: fixed;
  left: 0px;
  top: 0px;
  z-index: 1;
}

.reportTableForm input[readonly] {
  background-color: #f3f4f6; /* gray-100 */
```

```
  cursor: not-allowed;
  border-color: transparent;
  box-shadow: none;
}

.reportGenerateMain input,
.reportGenerateMain select {
  font-size: 14px;
  color: #000;
  height: 40px;
  line-height: 40px;
}

.colSpaceBottom > div {
  margin-bottom: 25px;
}

.page-break {
  page-break-before: always;
  break-before: page;
}

textarea {
  white-space: pre-wrap;
  word-wrap: break-word;
}

.banner {
```

```
  position: relative; /* Make sure the parent element is
positioned */
  width: 100%;
  height: 100%;
  /* background: rgba(0, 0, 0, 0.3); */
}
.banner:before {
  content: "";
  width: 100%;
  height: 100%;
  background: rgba(0, 0, 0, 0.6);
  position: absolute;
  left: 0px;
  top: 0px;
  z-index: 10;
}

.banner:after {
  content: "";
  position: absolute;
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
  background: linear-gradient(to right, rgba(0, 0, 0, 0.6) 3%,
transparent 80%);
  z-index: 1; /* Lower than .block content */
}
```

```
.powerfulDataBox ul li {
  margin-top: 20px;
  margin-bottom: 25px;
}

.powerfulDataBox ul li h4 {
  font-weight: bold;
}

span#backButton:before {
  content: "";
  border-left: 1px solid #000;
  border-bottom: 1px solid #000;
  height: 10px;
  width: 10px;
  display: inline-block;
  transform: rotate(42deg);
  top: 0px;
  position: static;
  margin-right: 2px;
  margin-top: 10px;
}

.cardLoader {
  position: absolute;
  left: 0px;
  top: 0px;
```

```
  height: 100%;
  width: 100%;
  display: flex;
  justify-content: center;
  align-items: center;
}
.cardLoader img {
  width: 110px;
}

.filterCol .react-datepicker-wrapper {
  font-size: 16px;
  padding: 0px;
}

.filterCol input {
  padding: 5px;
  border-radius: 0px;
  border: 0px;
  border-bottom: 1px solid #000;
  background: transparent;
  margin-bottom: 5px;
}

.react-datepicker-wrapper {
  display: inline-block;
  width: 100%;
}
```

```
.react-datepicker__input-container input {
  width: 100%;
}

.filterDropDown:before {
  content: "";
  height: 0px;
  width: 0px;
  border-width: 10px;
  position: absolute;
  border-color: transparent transparent #f6f6f6;
  right: 38px;
  top: -20px;
  border-style: solid;
}

.filterDropDown label {
  margin-right: 5px;
  margin-top: 5px;
  min-width: 100px;
  text-align: right;
}

.filterDropDown input {
  border: 0px;
  border-bottom: 1px solid #ccc;
  border-radius: 0px;
  background: transparent;
```

```
}

.filterDropDown input:focus-visible {
  outline: none;
}

.datePickerCol {
  height: 35px;
}

.datePickerCol input.inputbox {
  background: url("../src/images/calenderIcon1.png") 97%
center no-repeat;
  background-size: 18px;
}

/* css for report tab by saurabh */
.globleFilter:focus-visible {
  outline: none;
  box-shadow: none;
}

.p-filter-column .p-sortable-column {
  background-color: #949090;
}
.p-datatable .p-sortable-column.p-highlight:focus {
  background: #949090;
  color: rgb(55, 65, 81);
```

```
}
.p-datatable .p-sortable-column.p-highlight {
  background: #949090;
  color: rgb(55, 65, 81);
}
.p-datatable .p-sortable-column.p-highlight .p-sortable-column-
icon {
  color: #9b4b90;
}
.p-button:focus {
  outline: none !important;
  box-shadow: none !important;
}

.leftMenuCol ul li {
  padding: 20px;
  color: #fff;
  position: relative;
}

.leftMenuCol ul li.selectedCol {
  background: #9b4b90;
}
.leftMenuCol ul li:hover {
  background: #9b4b90;
}
/* .leftMenuCol ul li.selectedCol:after {
  content: "";
```

```
  position: absolute;
  border-width: 33px;
  border-color: transparent transparent transparent #9b4b90;
  right: -66px;
  top: 0px;
} */

.hideDesktop {
  display: none;
}

.hideMobile {
  display: block;
}

.footerBox {
  background-color: #fff;
  padding-bottom: 0;
  margin-top: 0;
  max-width: 100%;
}

.footerFormBox {
  max-width: 1252px;
  padding: 0px 10px;
  color: #000;
  margin: 36px auto 20px;
  display: flex;
```

```
  justify-content: space-around;
}

.phonCol .footerLocationIcon img {
  height: 25px;
  width: auto;
  margin-top: 7px;
  margin-right: 5px;
}

.emailCol .footerLocationIcon img {
  height: 21px;
  width: 42px;
  max-width: inherit;
  margin-top: 5px;
  margin-right: 5px;
}

.footerInputBox {
  margin-top: 10px;
  margin-bottom: 10px;
}

.socialMediaLink {
  margin-top: 10px;
}

.footerInputBox.addressCol img {
```

```
  height: 38px;
  width: auto;
  min-width: 46px;
  margin-top: 3px;
}

p.footerLocation {
  color: #414141;
}

.addressCol,
.emailPhoneInfo {
}

.footerFormBox > div {
  width: 33.33%;
  margin-bottom: 30px;
  padding-left: 80px;
}

.footerSocalMediaIcon {
  margin-top: 10px;
}

.footerSocalMediaIcon a {
  margin-right: 10px;
  opacity: 0.2;
}
```

```
.footerSocalMediaIcon a:hover {
  opacity: 1;
}

.firstRowFC1,
.firstRowFC2,
.secondRowFC1,
.secondRowFC2 {
  width: 50%;
  padding: 0 20px;
  margin-bottom: 20px;
}

.thirdRow {
  width: 100%;
  padding: 0 20px;
  margin-bottom: 20px;
}

.secondRowFC1,
.selectInputCol {
  position: relative;
}

.selectInputCol {
  padding-left: 0;
}
```

```
.footerInputBox {
  margin-top: 10px;
  margin-bottom: 10px;
  display: flex;
}

.footerLocationIcon img {
  padding-right: 8px;
}

.footerEmail,
.footerPhone {
  color: #aa72ce;
  font-size: 17px;
  font-weight: 400;
}

.footerLocationTitle,
.footerSocalMediaTitle {
  color: #aa72ce;
  font-size: 20px;
  font-weight: 500;
}

.subFooterBox p {
  text-align: center;
  color: #414141;
```

```
  font-size: 16px;
  padding: 15px 0;
  background: #f1f1f1;
}

.scroll-to-top {
  position: fixed;
  bottom: 20px;
  right: 20px;
  background: #fff;
  color: #9b4b90;
  padding: 10px 15px;
  border-radius: 50%;
  cursor: pointer;
  box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
}

.scrollTopButton {
  width: 48px;
  text-align: center;
  padding: 7px 4px;
  line-height: 17px;
  font-size: 14px;
}

.chart-container {
  width: 100%;
  max-width: 1200px; /* or whatever max width you want */
```

```
  margin: 0 auto;
}

@keyframes blink {
  0%,
  100% {
    opacity: 1;
  }
  50% {
    opacity: 0.3;
  }
}

.blink {
  animation: blink 1s infinite;
}

@keyframes blinkn {
  0% {
    box-shadow: 0 0 0 0 rgba(22, 163, 74, 0.7);
  }
  70% {
    box-shadow: 0 0 0 10px rgba(22, 163, 74, 0);
  }
  100% {
    box-shadow: 0 0 0 0 rgba(22, 163, 74, 0);
  }
}
```

```
.blinkn {
  animation: blinkn 1.5s infinite;
}

.singnalRow {
  position: relative;
}
.singnalRow:before {
  content: "";
  width: 100%;
  height: 2px;
  position: absolute;
  left: 0px;
  bottom: 0px;
  background: linear-gradient(
    90deg,
    white 0%,
    rgba(220, 208, 219, 0.5) 50%,
    white 100%
  );
}

.p-toast-message {
  margin-left: 31px;
  font-size: 13px;
}
```

```
.p-datatable-table thead tr th {
  background: #9b4b90;
  color: white;
}

.p-datatable-table thead tr th svg {
  color: #fff;
}

.p-datatable-table thead tr:nth-child(2) {
  display: none;
}

.p-datatable-wrapper {
  border-radius: 10px 10px 0px 0px;
}

.p-datatable-wrapper th {
  font-weight: normal;
  font-size: 14px;
  border-right: 1px solid #752f6b;
}

.p-datatable .p-sortable-column.p-highlight,
.p-datatable .p-sortable-column.p-highlight:focus {
  background: #9b4b90;
  color: #fff;
}
```

```
.p-datatable .p-sortable-column.p-highlight .p-column-title {
  border-bottom: 1px solid #fff;
}

.p-datatable .p-sortable-column.p-highlight .p-sortable-column-
icon {
  color: #fff;
}

.p-datatable-wrapper td {
  font-size: 13px;
  border: 1px solid #ccc;
}

span.p-paginator-pages .p-highlight {
  background: #efefef;
  border-radius: 100%;
}

.popUpRow th:nth-child(1) {
  border-radius: 10px 0px 0px 0px;
}
.popUpRow th:last-child {
  border-radius: 0px 10px 0px 0px;
  border-right: 0px;
}
```

```
.popUpRow tr:nth-child(even) {
  background: #f8f8fa;
}

@media (max-width: 1540px) {
  .dashboardCard {
    max-width: 32%;
  }
}

@keyframes point-left-right {
  0%,
  100% {
    transform: translateX(0);
  }
  25% {
    transform: translateX(-5px); /* Move left */
  }
  50% {
    transform: translateX(5px); /* Move right */
  }
  75% {
    transform: translateX(-3px); /* Slight return */
  }
}

.pointer-animation {
  display: inline-block;
```

```
  animation: point-left-right 1s infinite;
}

@media (max-width: 1250px) {
  .colSpaceBottom > div {
    flex-direction: column;
  }
  .colSpaceBottom > div label {
    width: 100%;
    text-align: left;
    margin-bottom: 4px;
  }
}

@media (max-width: 1200px) {
  .filterRow {
    flex-wrap: wrap;
  }
  .filterButtonCol {
    width: 100%;
  }
  .datePickerCol,
  .lpCmsCol {
    width: 50%;
  }
  .lpCmsCol label {
    text-align: left;
  }
```

```
.filterButtonCol a {
  width: auto;
}

.dashboardCard {
  max-width: 49%;
}

.locoPilotTitle {
  flex-direction: column;
  border-bottom: 0px;
}
.locoPilotTitle .datePickerCol {
  width: 100%;
}
.locoPilotTitle span {
  border-bottom: 1px solid #ccc;
  margin-bottom: 10px;
  width: 100%;
}

.manageLocoPilotTitle {
  border-bottom: 0px;
  flex-direction: column;
}
.manageLocoPilotTitle span {
  border-bottom: 1px solid #ccc;
  margin-bottom: 10px;
```

```
  width: 100%;
  display: blockk;
}
.manageLocoPilotTitle .datePickerCol {
  width: 100%;
}
.manageLocoPilotTitle .buttonCol {
  width: auto;
}

.createFormRow {
  flex-direction: column;
}

.createFormColFirst {
  width: 100%;
}
.createFormColSecond {
  max-width: inherit;
  width: 100%;
}
.viewDownloadSection {
  position: static;
  margin-top: 30px;
  flex-wrap: wrap;
}
.createTemplateCol {
  flex-direction: column;
```

```
  }

  .createTemplateCol input {
    width: 100%;
  }
}

@media (max-width: 1060px) {
  .dailyReportTitle {
    border-bottom: 0px;
    flex-direction: column;
  }
  .dailyReportTitle span {
    border-bottom: 1px solid #ccc;
    width: 100%;
    display: block;
    margin-bottom: 10px;
  }
  .loginLeft {
    display: none;
  }
  .loginRight {
    display: flex;
    justify-content: center;
    width: 100%;
    padding-right: 0px;
    max-width: 100%;
  }
```

```
}
@media (max-width: 980px) {
  .bannerTitle h1 {
    font-size: 35px;
  }

  .formBox {
    display: flex;
    flex-direction: column;
  }

  .filterDropDown span {
    width: 100%;
    flex-wrap: wrap;
  }

  .lpCmsCol,
  .datePickerCol {
    width: 50%;
  }
  .filterButtonCol {
    margin: 20px 0px;
  }
  .datePickerCol input.inputbox {
    padding-right: 27px;
  }
  .emailPhoneInfo {
    padding-left: 0px !important;
```

```
  }

  .datePickerCol .searchCol {
    width: 100%;
  }
  .datePickerCol {
    flex-direction: column;
    height: auto;
  }

  .datePickerCol label {
    margin-bottom: 5px !important;
  }

  .datePickerCol .searchCol {
    padding-left: 0px;
    margin-top: 15px;
  }
  .datePickerCol input {
    width: 100%;
  }

  .manageLocoPilotTitle .datePickerCol .searchCol {
    margin-top: 3px;
  }
}

@media (max-width: 767px) {
```

```
.breakzoneBox,
.speedTimeGraph,
.drivingPatternBox {
  display: flex;
  flex-direction: column;
}
.speedTimeBanner {
  order: -1;
  padding-left: 0px;
}
.drivingPatternBox ul {
  flex-direction: column;
}
.drivingPatternBox li {
  margin-left: 0px;
  margin-right: 0px;
  margin-bottom: 10px;
}
.banner {
  display: none;
}
.bannerTitle {
  position: static;
  color: #000;
  background: #f1f1f1;
}
.bannerTitle h1 {
  color: #000;
```

```
  margin-top: 50px;
}

.logoHeader {
  height: 70px;
}
.logoText {
  font-size: 18px;
}
.containerLayout {
  margin-top: 45px;
}
.navIcon {
  display: flex;
}
.navList {
  position: absolute;
  right: 0px;
  width: 230px;
  background: #fff;
  --tw-shadow: 0 4px 6px -1px rgb(0 0 0 / 0.1),
    0 2px 4px -2px rgb(0 0 0 / 0.1);
  --tw-shadow-colored: 0 4px 6px -1px var(--tw-shadow-
color),
    0 2px 4px -2px var(--tw-shadow-color);
  box-shadow: var(--tw-ring-offset-shadow, 0 0 #0000),
    var(--tw-ring-shadow, 0 0 #0000), var(--tw-shadow);
  z-index: 40;
```

```
  border-radius: 5px;
  top: 45px;
}
.navList:before {
  content: "";
  height: 0px;
  width: 0px;
  border-width: 10px;
  border-color: transparent transparent #f1f1f1 transparent;
  position: absolute;
  top: -20px;
  right: 7px;
}

.filterDropDown {
  width: 100%;
}

.filterDropDown label {
  text-align: left;
}

.filterDropDown span {
  flex-direction: column;
}
.lpCmsCol,
.datePickerCol {
  width: 100%;
```

```
}
.datePickerCol input.inputbox {
  width: 100%;
}

.datePickerCol {
  width: 100%;
  margin-bottom: 20px;
  margin-top: 20px;
}

.hideDesktop {
  display: block;
}

.hideMobile {
  display: none;
}
.leftColInner {
  position: relative;
}

.leftMenuCol {
  height: inherit;
  position: static;
  width: 100%;
  margin-bottom: 5px;
}
```

```
.showSelectedItem {
  border-bottom: 1px solid #fff;
  padding: 20px;
  color: #fff;
  cursor: pointer;
  position: relative;
  font-size: 20px;
}
.showSelectedItem:after {
  content: "";
  position: absolute;
  height: 15px;
  width: 15px;
  border-left: 2px solid #fff;
  border-bottom: 2px solid #fff;
  right: 25px;
  transform: rotate(-45deg);
}

.showSelectedItem.active:after {
  transform: rotate(135deg);
  top: 30px;
}

.controlCenterCol {
  flex-direction: column;
  padding: 10px;
```

```
}

.rightContentCol {
  padding-left: 0px;
}

.leftMenuCol ul li.selectedCol:after {
  display: none;
}

.leftMenuCol ul {
  position: absolute;
  width: 100%;
  top: 71px;
}

.footerFormBox {
  flex-direction: column;
}

.footerFormBox > div {
  width: 100%;
}

.aboutUsContentBox {
  max-width: 92%;
}
```

```
.footerInputBox,
.footerInputBox p {
  flex-direction: column;
  text-align: center;
}

.footerInputBox p {
  display: flex;
  justify-content: center;
  align-items: center;
  max-width: 100%;
}

.socialMediaLink {
  text-align: center;
}

.footerSocalMediaIcon {
  display: flex;
  justify-content: center;
  align-items: center;
}

.footerFormBox > div:nth-child(3) {
  padding-left: 0;
}
.footerFormBox > div {
  padding-left: 0px;
```

```
}
.footerFormBox {
  max-width: 300px;
  margin: 20px auto 0;
}
.footerInputBox p {
  font-size: 15px;
}
.addressCol,
.emailPhoneInfo {
  position: relative;
  background: none;
  padding-bottom: 20px;
}

.addressCol:before,
.emailPhoneInfo:before {
  position: absolute;
  height: 1px;
  width: 100%;
  content: "";
  background: linear-gradient(90deg, #fff, #ccc 50%, #fff);
  bottom: -6px;
  left: 0;
}

.addressCol > div p:nth-child(3):before {
  left: 0;
```

```
  }

  .addressCol > div p:nth-child(3) {
    padding-top: 45px;
  }

  .addressCol > div p:nth-child(3):before {
    width: 100%;
    background-size: 28px;
    background-position: 50%;
  }

  .footerBox {
    max-width: 100%;
  }
  .dashboardCard {
    max-width: 100%;
  }
  .dashboardMainCol {
    padding-top: 20px;
  }
}

@media (max-width: 600px) {
  .bannerTitle h1 {
    font-size: 25px;
    line-height: 35px;
    margin-top: 0px;
```

```
}
.bannerTitle {
  padding: 40px;
}
.speedTimeGraph > div {
  width: 100%;
}
.insightFlSpeedCol ul {
  flex-direction: column;
}
.insightFlSpeedCol ul li {
  flex-direction: column;
  width: 100%;
}
.breakzoneBox > div {
  width: 100%;
  padding-left: 0px;
}
.insightFlSpeedCol h3 {
  font-size: 22px;
}
.breakzoneBox h3 {
  font-size: 22px;
  margin-top: 20px;
}
.drivingPatternBox h3 {
  font-size: 22px;
}
```

```
.logo {
  width: 220px;
}
.headerUserIcon {
  padding-right: 0px;
}
.speedTimeGraph h4 {
  font-size: 20px;
}

.insightFlSpeedCol li h4 {
  font-size: 20px;
  margin-top: 5px;
}

.drivingPatternBox h4 {
  font-size: 18px;
}

.dashboardCard {
  min-width: 100%;
}
span#backButton:before {
  margin-top: 6px;
  height: 7px;
  width: 7px;
}
.createReportForm > div {
```

```
  display: flex;
  flex-direction: column;
}
.createReportForm > div label {
  width: 100%;
  text-align: left;
  margin-right: 0px;
  font-size: 14px;
}
.createReportForm > div input,
.createReportForm > div select {
  width: 100%;
}
.createReportForm > div.goodColum input {
  width: auto;
}
.createReportForm > div.goodColum {
  display: inline-block;
  text-align: left;
}
.dragDropCol > div {
  width: 100%;
}

td::before {
  content: attr(data-label);
  font-weight: bold;
  display: block;
```

```
  margin-bottom: 5px;
  color: #ddd;
}
.responsiveTable tr {
  padding: 10px;
}
.dashboardCardOnlyRes ul li {
  display: flex;
  flex-direction: column;
}
.filterDropDown:before {
  right: 115px;
}

.reportViewTitle {
  text-align: left;
  padding-top: 50px;
  margin-top: 0px;
}
.reportViewTitle button {
  top: 48px;
}
.dateRangeCal {
  right: -20px;
}
.datePickerCol,
.lpCmsCol {
  display: flex;
```

```
  flex-direction: column;
  height: auto;
}
.lpCmsCol input {
  width: 100%;
}
.datePickerCol {
  margin: 5px 0px 10px 0px;
}
.filterDropDown label {
  position: relative;
}
.searchCol {
  margin-top: 10px;
  padding-left: 0px;
}
.searchCol input {
  width: 100%;
}
.showSelectedItem {
  padding: 10px 20px;
}
.leftMenuCol ul {
  top: 52px;
}
.showSelectedItem.active:after {
  top: 23px;
}
```

```
.showSelectedItem {
  font: 18px;
  padding-top: 10px;
  padding-bottom: 10px;
}
.leftMenuCol ul li {
  padding: 10px 20px;
}
.dailyReportTitle .datePickerCol {
  flex-direction: row;
}
.dailyReportTitle .datePickerCol label {
  margin-top: 2px;
}

.manageLocoPilotTitle .buttonCol {
  padding-left: 0px;
}

.manageLocoPilotTitle button {
  font-size: 13px;
}

ul.dashboardCardUi li span {
  text-align: left;
}
.createFormColFirst,
.createFormColSecond {
```

```
  display: flex;
  flex-direction: column;
  align-items: flex-start;
}
.goodColum input {
  max-width: 20px;
}
.createFormColFirst input {
  margin-left: 0px;
}
.createFormRow .dragDropCol {
  flex-direction: column;
  width: 300px;
}
.dropzondRow {
  width: 100%;
}
.createReportForm > div .viewDownloadSection label {
  text-align: center;
}
.viewDownloadSectionMain {
  width: 100%;
  max-width: 100%;
}

.createTemplateCol > div {
  flex-direction: column;
}
```

```
  .createTemplateCol > div label {
    text-align: left;
    width: 100%;
    padding-left: 10px;
  }
  .createTemplateCol .titleInput {
    width: 100%;
  }
}
/* small screen */
@media (max-width: 310px) {
  .filterButtonCol {
    flex-wrap: wrap;
  }

  .filterButtonCol button {
    margin-top: 10px;
  }
}

/**page not found css****/

.st0 {
  fill: #fff;
}
.st2 {
  fill: #5d89af;
}
```

```
.st3 {
  fill: #709abf;
}
.st4,
.st6 {
  fill: #fff;
  stroke: #b3dcdf;
  stroke-miterlimit: 10;
}
.st6 {
  stroke: #5d89af;
  stroke-width: 2;
}
.st7,
.st8,
.st9 {
  stroke: #709abf;
  stroke-miterlimit: 10;
}

.st7 {
  stroke-width: 5;
  stroke-linecap: round;
  fill: none;
}
.st8,
.st9 {
  fill: #fff;
```

```
}
.st9 {
  fill: none;
}
.st10 {
}

#cloud1 {
  animation: cloud003 15s linear infinite;
}

#cloud2 {
  animation: cloud002 25s linear infinite;
}

#cloud3 {
  animation: cloud003 20s linear infinite;
}

#cloud4 {
  animation: float 4s linear infinite;
}

#cloud5 {
  animation: float 8s linear infinite;
}

#cloud7 {
```

```
  animation: float 5s linear infinite;
}

#tracks {
  animation: slide 650ms linear infinite;
}

#bumps {
  animation: land 10000ms linear infinite;
}

@keyframes jig {
  0% {
    transform: translateY(0px);
  }
  50% {
    transform: translateY(1px);
  }
  100% {
    transform: translateY(0px);
  }
}

#car-layers {
  animation: jig 0.35s linear infinite;
}

@keyframes land {
```

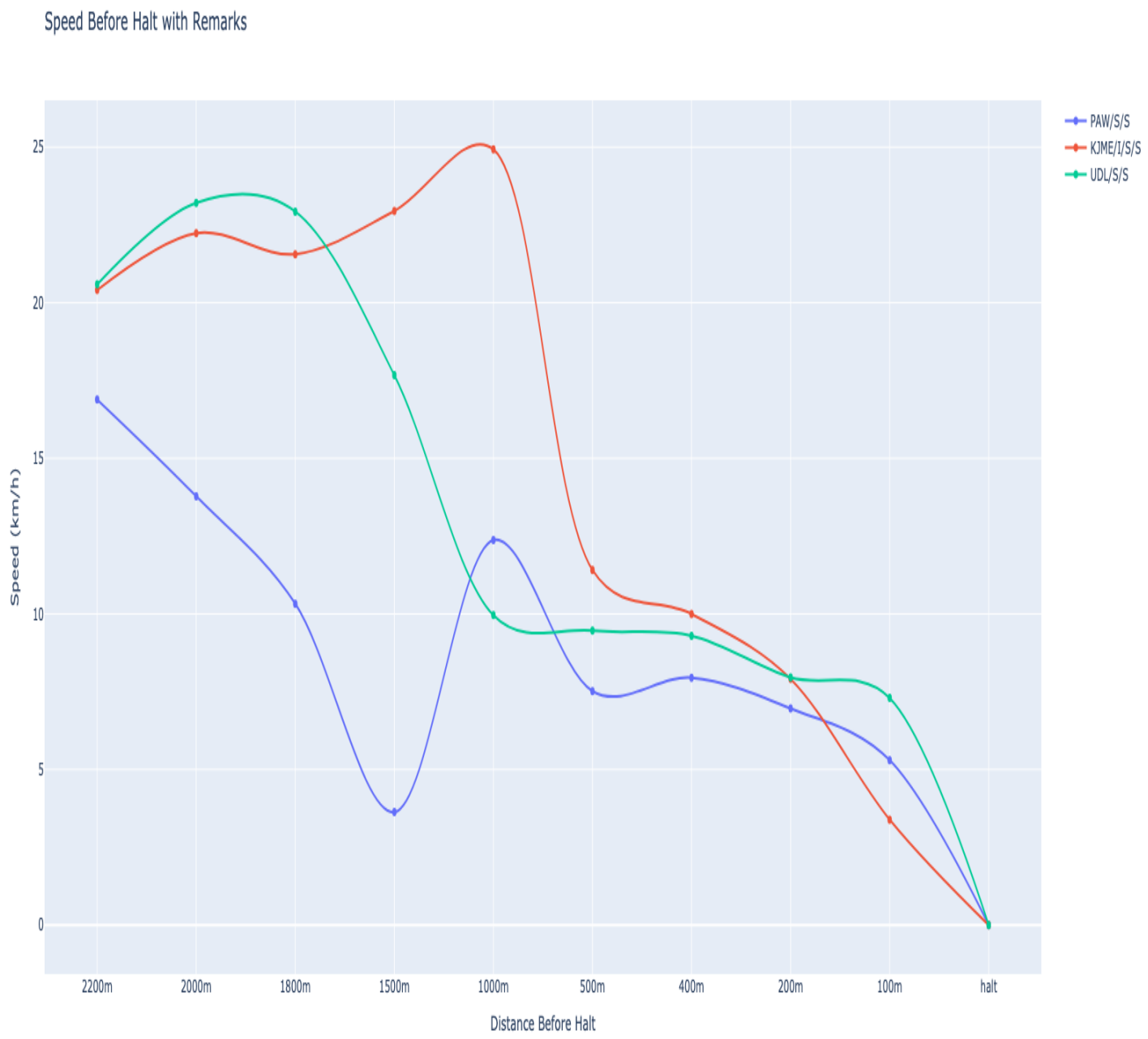
```
  from {
    transform: translateX(0);
  }
  to {
    transform: translateX(1000px);
  }
}

@keyframes slide {
  from {
    transform: translateX(0px);
  }
  to {
    transform: translateX(100px);
  }
}

/* @keyframes cloudFloat {
  0% { transform: translateX(0) translateY(3px); }
  100% { transform: translateX(1000px) translateY(0); }
} */

@keyframes cloud001 {
  0% {
    transform: translateX(-1000px) translateY(3px);
  }
  100% {
    transform: translateX(1000px) translateY(0);
```

```
  }
}

@keyframes cloud002 {
  0% {
    transform: translateX(-1000px) translateY(3px);
  }
  100% {
    transform: translateX(1000px) translateY(0);
  }
}

@keyframes cloud003 {
  0% {
    transform: translateX(-1000px) translateY(3px);
  }
  100% {
    transform: translateX(1000px) translateY(0);
  }
}

@keyframes float {
  0% {
    transform: translateY(0px) translateX(0);
  }
  50% {
    transform: translateY(8px) translateX(5px);
  }
```

```
  100% {
    transform: translateY(0px) translateX(0);
  }
}

#bracefront,
#braceback {
  animation: braces 1s linear infinite;
}

@keyframes braces {
  0% {
    transform: translateX(-2px);
  }
  25% {
    transform: translateX(3px);
  }
  50% {
    transform: translateX(-2px);
  }
  75% {
    transform: translateX(3px);
  }
  100% {
    transform: translateX(-2px);
  }
}
```

```
.p-dropdown:focus,
.p-dropdown:focus-visible {
  outline: none;
  box-shadow: none;
  border-color: transparent;
}

.p-dropdown {
  height: 42px; /* or your desired height */
  padding: 8px 6px;
  font-size: 14px;
}

.p-dropdown-label {
  line-height: 1.2;
  padding: 2px 0;
}

.p-dropdown:focus-within {
  outline: none;
  box-shadow: none;
  border-color: #e5e7eb; /* or your desired color */
}
.p-toast.p-component.p-toast-top-right.p-ripple-disabled {
    margin-top: 80px;
}
```

Generated speed breaking patter



CODE :

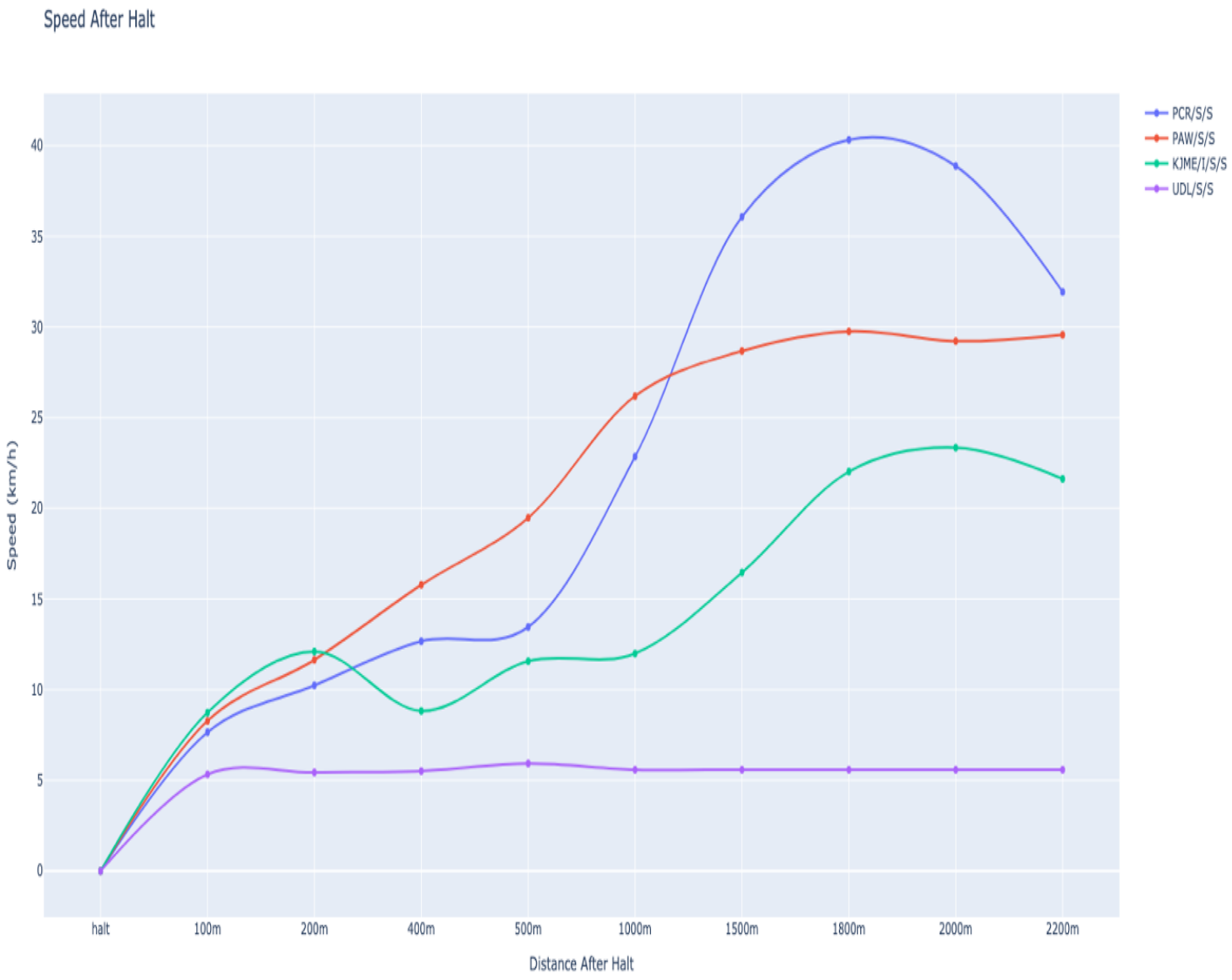
```
def speed_before_halt(report_id):
```

```
    logging.info(f'speed_before_halt ...')
    apiResponse = ApiResponse(200, 'ok', None)
    reportService = ReportService()
    try:
        claims = get_jwt()
        u_id = claims["u_id"]
        org_id = claims["org_id"]
        from_station = request.args.get('from_station')
        to_station = request.args.get('to_station')
        fig = reportService.speed_before_halt(report_id,
from_station, to_station, org_id)
        apiResponse.data = pio.to_json(fig)
    except RailGenericError as e:
        error_trace = traceback.format_exc()
        logging.info(f'Add report failed. error :: {e}. error_trace ::
{error_trace}')
        apiResponse.status = e.code
        apiResponse.message = str(e)
    except Exception as e:
        error_trace = traceback.format_exc()
        logging.info(f'add report failed. error :: {e}. error_trace ::
{error_trace}')
        apiResponse.status = 500
        apiResponse.message = ERR_MSG.UNKNOWN

    return jsonify(apiResponse.__dict__)

@bp.route('/<int:report_id>/stat_speed_before_halt',
```

```
methods=['GET'])
@jwt_required()
def stat_speed_before_halt(report_id):
    logging.info(f'stat_speed_before_halt ...')
    apiResponse = ApiResponse(200, 'ok', None)
    reportService = ReportService()
    try:
        claims = get_jwt()
        u_id = claims["u_id"]
        org_id = claims["org_id"]
        from_station = request.args.get('from_station')
        to_station = request.args.get('to_station')
        df = reportService.stat_speed_before_halt(report_id,
from_station, to_station,org_id)
        apiResponse.data = df.to_dict(orient='records')
    except RailGenericError as e:
        error_trace = traceback.format_exc()
        logging.info(f'Add report failed. error :: {e}. error_trace ::
{error_trace}')
        apiResponse.status = e.code
        apiResponse.message = str(e)
    except Exception as e:
        error_trace = traceback.format_exc()
        logging.info(f'add report failed. error :: {e}. error_trace ::
{error_trace}')
        apiResponse.status = 500
        apiResponse.message = ERR_MSG.UNKNOWN
```

```
    return jsonify(apiResponse.__dict__)

@bp.route('/<int:report_id>/speed_after_halt',
methods=['GET'])
@jwt_required()
```

Generated acceleration pattern



Code :

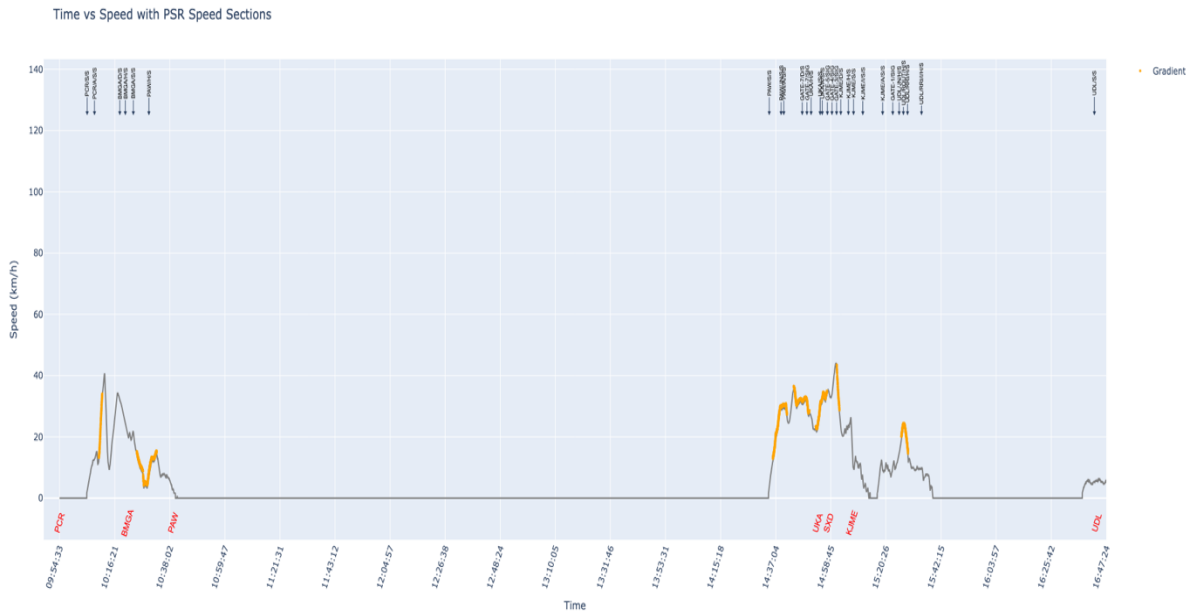
```
def speed_after_halt(report_id):
    logging.info(f'speed_after_halt ...')
    apiResponse = ApiResponse(200, 'ok', None)
    reportService = ReportService()
```

```
    try:
        claims = get_jwt()
        u_id = claims["u_id"]
        org_id = claims["org_id"]
        from_station = request.args.get('from_station')
        to_station = request.args.get('to_station')
        fig = reportService.speed_after_halt(report_id,
from_station, to_station, org_id)
        apiResponse.data = pio.to_json(fig)
    except RailGenericError as e:
        error_trace = traceback.format_exc()
        logging.info(f'Add report failed. error :: {e}. error_trace ::
{error_trace}')
        apiResponse.status = e.code
        apiResponse.message = str(e)
    except Exception as e:
        error_trace = traceback.format_exc()
        logging.info(f'add report failed. error :: {e}. error_trace ::
{error_trace}')
        apiResponse.status = 500
        apiResponse.message = ERR_MSG.UNKNOWN

    return jsonify(apiResponse.__dict__)

@bp.route('/<int:report_id>/stat_speed_after_halt',
methods=['GET'])
```

```
@jwt_required()
def stat_speed_after_halt(report_id):
    logging.info(f'stat_speed_after_halt ...')
    apiResponse = ApiResponse(200, 'ok', None)
    reportService = ReportService()
    try:
        claims = get_jwt()
        u_id = claims["u_id"]
        org_id = claims["org_id"]
        from_station = request.args.get('from_station')
        to_station = request.args.get('to_station')
        df = reportService.stat_speed_after_halt(report_id,
from_station, to_station, org_id)
        apiResponse.data = df.to_dict(orient='records')
    except RailGenericError as e:
        error_trace = traceback.format_exc()
        logging.info(f'Add report failed. error :: {e}. error_trace ::
{error_trace}')
        apiResponse.status = e.code
        apiResponse.message = str(e)
    except Exception as e:
        error_trace = traceback.format_exc()
        logging.info(f'add report failed. error :: {e}. error_trace ::
{error_trace}')
        apiResponse.status = 500
        apiResponse.message = ERR_MSG.UNKNOWN
```

```
    return jsonify(apiResponse.__dict__)

@bp.route('/', methods=['POST'])
@jwt_required()
```

Generated Speed time chat:



CODE :

```
def chart_speed_time_json(report_id):
    logging.info(f'chart_speed_time ...')
    apiResponse = ApiResponse(200, 'ok', None)
    reportService = ReportService()
    try:
        claims = get_jwt()
        u_id = claims["u_id"]
        org_id = claims["org_id"]
        from_station = request.args.get('from_station')
        to_station = request.args.get('to_station')
        print(from_station, to_station)
        fig = reportService.generate_speed_time_chart(report_id,
from_station, to_station, org_id)
        apiResponse.data = pio.to_json(fig)
    except RailGenericError as e:
        error_trace = traceback.format_exc()
        logging.info(f'Add report failed. error :: {e}. error_trace ::
{error_trace}')
        apiResponse.status = e.code
        apiResponse.message = str(e)
    except Exception as e:
        error_trace = traceback.format_exc()
```

```
        logging.info(f'add report failed. error :: {e}. error_trace ::
{error_trace}')
        apiResponse.status = 500
        apiResponse.message = ERR_MSG.UNKNOWN

    return jsonify(apiResponse.__dict__)

@bp.route('/<int:report_id>/speed_before_halt',
methods=['GET'])
@jwt_required()
```

TEMPLATE :

CODE :

```
import React, { useState, useEffect } from "react";
import { Link } from "react-router-dom";
import banner from "../../src/images/bannerRailwayN.png";
```

```
import sideBanner from
"../../src/images/railwaySideBanner.png";
import sideBanner2 from
"../../src/images/railwaySideBanner2.png";
import DemoRequestForm from
"../components/DemoRequestForm";
import { getDataFromLocalStorage } from
"../utils/localStorage";

const HomeComponent = () => {
  const [showDemoForm, setShowDemoForm] = useState(false);
  const [userInfo, setUserInfo] = useState(null);
  useEffect(() => {
    const user = getDataFromLocalStorage("userInfo");
    setUserInfo(user);
  }, []);

  return (
    <div>
      <div className="w-full flex items-center ">
        <div className="bannerTitle w-full block max-w-[1080px]
mx-auto  p-[50px] pt-[0px] text-white absolute z-[11]">
          <h1 className="text-[56px] pb-[15px] text-[#fff]">
            Powerful Data Analytics for Railways
          </h1>
          <p className="max-w-[675px] text-[17px] mb-[15px]">
```

```
          Analyze, improve, and guide loco pilots with real-time insights for
          safer, more efficient railway operations across India.
        </p>
        {!userInfo && (
          <div>
            <span
              onClick={() => setShowDemoForm(true)}
              id="getStartedButton"
              className=" cursor-pointer reportGenerateBg rounded-[4px] text-[#fff] sm:text-[16px] text-[14px] sm:px-[15px] px-[12px] py-[8px] sm:py-[15px] inline-block mt-[20px]"
            >
              REQUEST A DEMO
            </span>
          </div>
        )}
      </div>
      <div className="banner">
        <img src={banner} className="w-full" />
      </div>
    </div>
    <div className="w-full">
      <div className="max-w-[1200px] w-full mx-auto sm:my-[80px] my-[40px] px-4">
        <div className="w-full powerfulDataBox">
```

```
{/* <h3 className="text-[34px] font-semibold">
  Powerful Data Analytics for Railways
</h3> */}
<div className="w-full flex flex-row speedTimeGraph">
  <div className="w-[60%] pr-[20px]">
    <ul>
      <li>
        <h4 className="text-[24px] text-[#9b4b90]">
          Speed-Time Graphs
        </h4>
        <p>
          Visualize loco speed trends to monitor operational
          efficiency and punctuality with precise real-time
data
          tracking.
        </p>
      </li>
      <li>
        <h4 className="text-[24px] text-[#9b4b90]">
          Braking Profile Analysis
        </h4>
        <p>
          Analyze braking patterns to enhance safety and
optimize
          braking efficiency, reducing wear and increasing
train
```

```
                stability.
              </p>
            </li>
            <li>
              <h4 className="text-[24px] text-[#9b4b90]">
                Guideline Violation Detection
              </h4>
              <p>
                Detect overspeed, late braking, and other violations
                instantly to enforce compliance and prevent accidents.
              </p>
            </li>
          </ul>
        </div>
        <div className="w-[40%] speedTimeBanner pl-[20px]">
          <img src={sideBanner} className="w-full" />
        </div>
      </div>
    </div>
  </div>
</div>
<div className="w-full reportGenerateBg">
  <div className="max-w-[1200px] w-full mx-auto sm:my-
```

```
[80px] my-[40px] py-16 px-4">
        <div className="bg-white sm:my-4 my-0 p-2 rounded-
[10px] insightFlSpeedCol">
          <h3 className="text-[30px] text-center py-4 text-
[#9b4b90] font-semibold">
            Insightful Speed-Time Curve Comparison
          </h3>
          <div className="w-full">
            <ul className="px-2 flex flex-row justify-between max-
w-[980px] mx-auto my-8">
              <li className="flex flex-row mr-2 w-[40%] mb-4">
                <span className="w-[32px] mr-2">
                  <svg
                    className="colorable-icon w-[32px] h-[32px] mt-2
mr-4"
                    viewBox="0 0 512 512"
                    xmlns="http://www.w3.org/2000/svg"
                    data-icon="chart-line-up-down"
                    data-prefix="fal"
                    aria-hidden="true"
                  >
                    <path
                      d="M32 48c0-8.8-7.2-16-16-16S0 39.2 0 48V400c0
44.2 35.8 80 80 80H496c8.8 0 16-7.2 16-16s-7.2-16-16-16H80c-
26.5 0-48-21.5-48-48V48zM368 96h57.4L288 233.4l-68.7-68.7c-
3-3-7.1-4.7-11.3-4.7s-8.3 1.7-11.3 4.7l-96 96c-6.2 6.2-6.2 16.4 0
```

22.6s16.4 6.2 22.6 0L208 198.6l68.7 68.7c6.2 6.2 16.4 6.2 22.6 0L448 118.6V176c0 8.8 7.2 16 16 16s16-7.2 16-16V80c0-8.8-7.2-16-16-16H368c-8.8 0-16 7.2-16 16s7.2 16 16 16zm0 288h96c8.8 0 16-7.2 16-16V272c0-8.8-7.2-16-16-16s-16 7.2-16 16v57.4l-68.7-68.7-22.6 22.6L425.4 352H368c-8.8 0-16 7.2-16 16s7.2 16 16 16z"
fill="currentColor"
></path>
</svg>

<h4 className="text-[22px] font-semibold">
Before Training
</h4>

Irregular speed, sharp accelerations and braking reducing
efficiency.

</li>
<li className="flex flex-row mr-2 w-[40%] mb-4">

<svg
className="colorable-icon w-[32px] h-[32px] mt-2 mr-4"
viewBox="0 0 512 512"

```
                xmlns="http://www.w3.org/2000/svg"
                data-icon="chart-line-up"
                data-prefix="fal"
                aria-hidden="true"
              >
                <path
                  d="M32 48c0-8.8-7.2-16-16-16S0 39.2 0 48V400c0
44.2 35.8 80 80 80H496c8.8 0 16-7.2 16-16s-7.2-16-16-16H80c-
26.5 0-48-21.5-48-48V48zm288 96c0 8.8 7.2 16 16 16h89.4L320
265.4l-84.7-84.7c-6.2-6.2-16.4-6.2-22.6 0l-112 112c-6.2 6.2-6.2 16.4
0 22.6s16.4 6.2 22.6 0L224 214.6l84.7 84.7c6.2 6.2 16.4 6.2 22.6
0L448 182.6V272c0 8.8 7.2 16 16 16s16-7.2 16-16V144c0-8.8-7.2-16-
16-16H336c-8.8 0-16 7.2-16 16z"
                  fill="currentColor"
                ></path>
              </svg>
            </span>
            <div>
              <h4 className="text-[22px] font-semibold">
                After Training
              </h4>
              <p>
                Smoother driving with optimal speed control
enhancing
                safety and fuel economy.
              </p>
```

```
              </div>
            </li>
          </ul>
        </div>
      </div>
    </div>
  </div>
  <div className="w-full">
    <div className="max-w-[1200px] w-full mx-auto sm:my-
[80px] my-[40px] px-4">
      <div className="w-full powerfulDataBox">
        <div className="w-full flex flex-row breakzoneBox">
          <div className="w-[40%] pl-[20px]">
            <img src={sideBanner2} className="w-full" />
          </div>
          <div className="w-[60%] pl-[20px]">
            <h3 className="text-[34px] font-semibold text-
[#9b4b90]">
              Braking Zone Visual Analysis
            </h3>
            <ul>
              <li>
                <p>
                  This visual tool identifies critical braking zones,
                  flagging late applications and unnecessary early
brakes
```

```
                for targeted pilot training.
              </p>
            </li>
            <li>
              <p>
                Reducing braking inconsistencies ensures greater passenger
                comfort, less mechanical stress, and enhanced safety.
              </p>
            </li>
          </ul>
        </div>
      </div>
    </div>
  </div>
  <div className="w-full bg-[#efefef]">
    <div className="max-w-[1200px] w-full mx-auto sm:py-[80px] py-[40px] px-4">
      <div className="w-full p-2 drivingPatternBox">
        <h3 className="text-center text-[26px] mb-8 font-semibold">
          Driving Pattern Suggestions
        </h3>
        <ul className="flex flex-row justify-bweteen">
```

```
<li class=" bg-white p-4  rounded-[5px]">
  <span>
    <h4 class="text-[20px] font-semibold">
      Optimized Speed Control
    </h4>
    <p class="text-[#434343]">
      Encourages steady acceleration to reduce energy consumption.
    </p>
  </span>
</li>
<li class=" bg-white p-4 mx-4 rounded-[5px]">
  <span>
    <h4 class="text-[20px] font-semibold">Timely Braking</h4>
    <p class="text-[#434343]">
      Promotes early and smooth braking for increased safety and
      efficiency.
    </p>
  </span>
</li>
<li class=" bg-white p-4  rounded-[5px]">
  <span>
    <h4 class="text-[20px] font-semibold">
      Compliance Reinforcement
```

```
                </h4>
                <p className="text-[#434343]">
                  Alerts on any deviations to ensure adherence to
safety
                  guidelines.
                </p>
              </span>
            </li>
          </ul>
        </div>
      </div>
    </div>
    {showDemoForm && (
      <DemoRequestForm onClose={() =>
setShowDemoForm(false)} />
    )}
  </div>
  );
};

export default HomeComponent;
```

REPORT ALLL SPEED CONTROL CODE :

```
def speed_after_halt(report_id):
    logging.info(f'speed_after_halt ...')
    apiResponse = ApiResponse(200, 'ok', None)
    reportService = ReportService()
    try:
        claims = get_jwt()
        u_id = claims["u_id"]
        org_id = claims["org_id"]
        from_station = request.args.get('from_station')
        to_station = request.args.get('to_station')
        fig = reportService.speed_after_halt(report_id,
from_station, to_station, org_id)
        apiResponse.data = pio.to_json(fig)
    except RailGenericError as e:
        error_trace = traceback.format_exc()
        logging.info(f'Add report failed. error :: {e}. error_trace ::
{error_trace}')
        apiResponse.status = e.code
        apiResponse.message = str(e)
    except Exception as e:
        error_trace = traceback.format_exc()
        logging.info(f'add report failed. error :: {e}. error_trace ::
{error_trace}')
        apiResponse.status = 500
        apiResponse.message = ERR_MSG.UNKNOWN
```

```
    return jsonify(apiResponse.__dict__)

@bp.route('/<int:report_id>/stat_speed_after_halt',
methods=['GET'])
@jwt_required()
def stat_speed_after_halt(report_id):
    logging.info(f'stat_speed_after_halt ...')
    apiResponse = ApiResponse(200, 'ok', None)
    reportService = ReportService()
    try:
        claims = get_jwt()
        u_id = claims["u_id"]
        org_id = claims["org_id"]
        from_station = request.args.get('from_station')
        to_station = request.args.get('to_station')
        df = reportService.stat_speed_after_halt(report_id,
from_station, to_station, org_id)
        apiResponse.data = df.to_dict(orient='records')
    except RailGenericError as e:
        error_trace = traceback.format_exc()
        logging.info(f'Add report failed. error :: {e}. error_trace ::
{error_trace}')
        apiResponse.status = e.code
        apiResponse.message = str(e)
    except Exception as e:
```

```
        error_trace = traceback.format_exc()
        logging.info(f'add report failed. error :: {e}. error_trace ::
{error_trace}')
        apiResponse.status = 500
        apiResponse.message = ERR_MSG.UNKNOWN

    return jsonify(apiResponse.__dict__)

@bp.route('/', methods=['POST'])
@jwt_required()
```

Future Scope :

1. Integration with Real-Time IoT Devices.
2. AI-Based Predictive Violation Detection.
3. Fuel Consumption Optimization Analytics.
4. Mobile Application for Railway Officers.
5. Integration with Central Railway Database.
6. Machine Learning-based Driver Performance Scoring.
7. Cloud Deployment (AWS/Azure).

REFERNCES :

The following books and resources were referred during the development of the RailPlot System:

Software Engineering: A Practitioner's Approach – by Roger S. Pressman

MySQL: The Complete Reference

Concurrent Programming in Java – by Doug Lea

Java Black Book – by Steven Holzner

ONLINE REFERENCES

Oracle Corporation Official Documentation (MySQL)

W3Schools – For SQL, HTML, and backend concepts

GeeksforGeeks – For programming and DFD concepts

GitHub – For project hosting and version control